

AI and the Ladder of Development

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Preliminary and incomplete draft for discussion only. This version presents some early descriptives and develops the theoretical/conceptual framework. Mistakes likely. Co-developed with Claude.

Abstract

Will artificial intelligence (AI) help countries ascend the ladder of development or will it remove the rungs? I develop a small open-economy model that allocates labour across manufacturing, progressive tradable services, and local non-tradable services. Productive capability governs both absorptive capacity and long-run learning.

Frontier AI raises domestic productivity in tradable sectors, intensifies foreign competition in those same sectors, and changes the learning content of the work itself. Its effect on capability growth decomposes into (i) an alignment term capturing the covariance between sectoral learning weights and AI-induced labour-demand effects, (ii) a Baumol reallocation term which is negative when AI raises tradable labour demand relative to local-service productivity, and (iii) a direct learning-content term capturing how AI changes what workers learn in the jobs that remain. AI can therefore raise domestic tradable-sector productivity and contemporaneous real income while lowering long-run capability growth, producing a potential development trap: wages, real income, and consumer-service employment all rise while the growth engine quietly contracts.

The paper works out when this trap is merely a growth cost and when it is welfare-reducing, shows how a slowly advancing AI frontier can generate discontinuous capability collapses and take-offs, and derives an optimal learning-weighted subsidy. This draft covers the theory, together with a first pass over public frontier-AI usage data from the Anthropic Economic Index which supplies descriptive counterparts for several of the model's objects (the cross-country gradient of usage intensity, the task composition of frontier traffic, and the composition of delegation) while leaving their structural magnitudes, and the learning-content parameters in particular, unidentified. Taking the model's sufficient statistics to data is ongoing.

1 Introduction

Will artificial intelligence (AI) help countries ascend the ladder of development or will it remove the rungs? Two opposing views dominate the debate. The optimistic view holds that AI is the

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ultimate leapfrog technology: frontier intelligence travels as software, so a country can import capability rather than build it, and can export “disembodied” services without ever industrialising [Bhagwati \(1984\)](#); [Baldwin \(2019\)](#). The pessimistic view holds that AI erodes exactly the rungs of the development ladder that carried past growth episodes: routine cognitive service exports (and, in time, manufacturing) are especially exposed, so AI extends the logic of premature deindustrialisation ([Rodrik, 2016](#)) from factories to the service ladder itself. Both views are claims about *composition*. The question is not the size of the average productivity gain; it is which activities AI pulls an economy’s labour force into.

This paper builds a simple open-economy general-equilibrium model in which that compositional question can be posed exactly and answered with a sufficient statistic. The economy has three sectors: manufacturing tradables (M), progressive tradable services (P), and consumer non-tradable services (C). Two state variables track the domestic side of the model: productive capability H , which evolves with the sectoral allocation of labour through learning weights $\psi_s(H)$, and an AI-readiness stock D , distinct from capability, which governs how much of the frontier a country implements. Frontier AI, indexed by an exogenous state F , affects the economy through three channels that operate on the same sectors and share a common origin in task automation: it raises domestic tradable productivity at a rate $\alpha_s(H, D)$, it intensifies foreign competition in those sectors at rate ν_s , and it changes the learning content of work itself. The same delegation of tasks that raises productivity can strip out, or enrich, the tasks through which workers accumulate capability, so the learning weights $\psi_s(H, F)$ respond to the frontier directly. Preferences make tradables and local services gross complements, so productivity gains in tradables endogenously pull labour into non-tradable services—the force of [Baumol \(1967\)](#) and [Ngai and Pissarides \(2007\)](#), which [Peters et al. \(2026\)](#) document as the dominant pattern of structural change in much of the developing world.

The framework is in the Hulten-style spirit of recent AI macroeconomics ([Acemoglu, 2025](#); [Jones, 2026](#)). For development, whether AI adds a percentage point to average TFP matters far less than where it moves people—and what they learn once there. The welfare, transition, and policy results below are consequences of a development multiplier, which I derive. I establish the following.

I. A decomposition of how advancements in AI shape capability growth. The effect of a marginal frontier advance on capability growth splits into (i) a covariance term between sectoral learning weights ψ_s and labour-demand effects ϑ_s , (ii) a Baumol term proportional to the average labour-demand gain, and (iii) a learning-content term that captures AI changing what workers learn in the jobs that remain. Average productivity effects are not sufficient statistics for development: an AI shock that raises tradable labour demand *on average* can strictly reduce capability growth if the gains are concentrated in the wrong tradable sector. An AI shock with favorable composition effects can still hollow out capability if it automates precisely the tasks through which workers learned. Thus, it establishes the leapfrog-versus-ladder debate as a measurable alignment condition. A third margin sits inside the learning-content term: whether AI arrives as automation or augmentation.

II. A potential development trap is an emergent property, not an assumption. In reduced form, whether AI raises sectoral demand for labour (ϑ_s positive or negative) turns on whether absorption

$(\mu - 1)\alpha_s$ beats foreign competition $\mu\nu_s$. A task-level microfoundation—home firms implement AI in a share $g(D)$ of automatable tasks, frontier competitors implement it in all of them, and world demand for the sector has price elasticity ζ_s —reveals what the reduced form hides: an interior threshold exists *only if* $\zeta_s > 1$. A further decomposition then separates what ϑ_s bundles: the frontier’s effect on home market share, which depends only on the absorption gap $1 - g(D)$ and which full absorption always neutralizes, and its effect on the size of the world market, which depends only on ζ_s . If world demand for a sector is price-inelastic, frontier AI reduces world expenditure on the sector no matter who absorbs it, so that even under full domestic absorption the home sector’s labour demand cannot rise. Absorption defends market share and output, but not employment. Whether AI can ever be an employment escalator in business services is thus a question about the world demand curve, and not only about domestic readiness.

III. The trap is invisible in aggregate indicators. In the trap region, AI raises average tradable labour demand but reduces capability growth. Wages and real income rise and consumer-service employment expands, yet the growth engine contracts. The damage is visible only in compositional data—sectoral employment, exports, market shares—and in capability-sensitive outcomes that respond with a lag. A closed-form present-value calculation then gives a first-order immiserization condition: around a stable steady state, AI lowers welfare in the trap if and only if $\beta|\mathcal{M}| > (\rho + \kappa)\bar{\vartheta}$, where β is the capability–wage feedback, κ the speed of convergence, and ρ the discount rate. The trap is a welfare trap for patient economies with strong capability–wage feedback, and “only” a growth trap otherwise.

IV. Continuous frontier progress generates discontinuous development outcomes. Along a slowly moving frontier the economy tracks a moving steady-state locus whose slope is the development multiplier itself. When the locus folds, the multiplier’s sign determines whether frontier progress carries the economy toward or away from the fold: a capability collapse (the escalator is removed under the economy’s feet) or a take-off wave (the frontier dissolves the trap). Both transitions are hysteretic: after a collapse, halting frontier progress does not restore the old steady state, and escape requires a discrete big push in the sense of [Murphy et al. \(1989\)](#).

V. The development multiplier is the policy-relevant externality. Because learning spillovers are uninternalized, the competitive equilibrium is inefficient by construction. The planner’s allocation is decentralized by ladder subsidies proportional to the shadow value of capability times sectoral learning weights, alongside a terms-of-trade export tax that vanishes in the empirically relevant atomistic limit. The model therefore does not rationalize export promotion per se; it rationalizes paying for learning, wherever learning occurs.

1.1 Related literature

The paper connects four literatures. The structural transformation literature ([Baumol, 1967](#); [Ngai and Pissarides, 2007](#); [Buera and Kaboski, 2012](#); [Herrendorf et al., 2014](#)) supplies the Baumol force; [Fan et al. \(2023\)](#) and [Peters et al. \(2026\)](#) document service-led structural change in developing economies and provide the empirical foil for my “growth without capability” region, and [Nordhaus](#)

(2008) provides the aggregate evidence for the demand-side Baumol mechanism. Relative to the service-led growth agenda, the model is explicit about which margin AI moves: the export demand facing modern services and their capacity to generate capability, rather than productivity within those services or access to modern-service employment. The trade-and-learning literature (Krugman, 1987; Young, 1991; Atkin et al., 2026) supplies the dynamic spillover object ψ_s and the idea that comparative advantage today shapes capability tomorrow; Rodrik (2016) is the empirical benchmark for trade-induced descent of the ladder, and Atkin et al. (2017) provide experimental evidence that learning attaches to export relationships. The technology diffusion literature motivates capability-dependent absorption: Comin and Mestieri (2018) show that adoption lags and penetration gaps between rich and poor countries account for a large share of income divergence, and Hjort and Poulsen (2019) provide microevidence that the arrival of fast internet in Africa raised employment—disproportionately in higher-skill occupations—alongside firm entry, productivity, and exporting. The AI macroeconomics literature provides discipline on magnitudes and mechanisms: Acemoglu (2025) and Jones (2026) on aggregate effects, Acemoglu and Restrepo (2018) on the task framework our microfoundation borrows, Aghion et al. (2019) on growth channels, and Korinek and Stiglitz (2021) and Alonso et al. (2020) on the specific concern—which this paper formalizes—that AI may erode developing countries’ comparative advantage and cause divergence. Korinek and Stiglitz (2021) also stress an international/domestic distinction that the policy section inherits. Finally, firm-level evidence that generative AI produces its largest productivity gains among less experienced workers (Brynjolfsson et al., 2025) motivates the learning-content channel: whether AI hollows out or accelerates on-the-job learning is an empirical question that can determine the sign of the development effect, so the baseline model carries it rather than assuming it away.

Relative to all of these, the contribution is a single object: the covariance-minus-Baumol-plus-learning-content decomposition of AI’s development effect, together with the microfoundation of its thresholds, its welfare content, its transition dynamics, and its policy counterpart.

2 Motivating facts from observed frontier-AI use

The model below is built around three features of AI diffusion: (i) that adoption depends on a readiness stock and is currently very unequal across countries; (ii) that frontier use is concentrated in task families common in digitally tradable services; and (iii) that how much is adopted and how it is *implemented* are distinct margins, so that the same technology can arrive as a tool in one place and as a substitute in another.

The Anthropic Economic Index (AEI) classifies conversations on Anthropic’s Claude chat products and traffic on its first-party (1P) API into occupational tasks (O*NET) and bottom-up task categories, and labels each exchange by collaboration mode: directive delegation or automation, in which the model executes the task outright, against augmentative modes in which the user iterates (Handa et al., 2025; Anthropic, 2025).

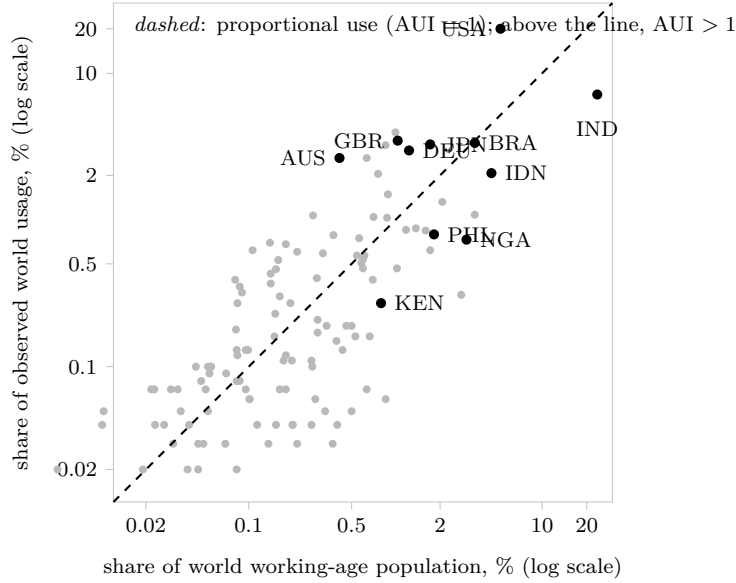


Figure 1: Scale against intensity: each country’s share of observed world usage against its share of world working-age population. Source: Anthropic (2026c).

Fact 1: frontier-AI use is extremely unequal across countries, and volume and intensity diverge. The Anthropic Usage Index (AUI) is a country’s share of observed Claude usage divided by its share of the world’s working-age population. The May 2026 aggregate points to considerable divergence, with rich countries exceeding ratios of 2–3x while developing countries sit well below 0.5. A one-percent higher GDP per capita is associated with 0.7 percent higher usage per capita across countries, and 1.8 percent across US states (Anthropic, 2025).

This is the fact that motivates a *readiness* stock entering the implementation share $g(D)$ of Section 3.5, rather than a uniform productivity shifter. This is also the reason readiness is carried as a state separate from productive capability. Productive capability accumulates through learning by doing and is slow-moving by construction, whereas the levels and the top of this ranking move substantially in eight months, and the index is shaped by payment access, language coverage, sanctions, user composition, and product mix—all of which can change when a product is launched, localized, repriced, or licensed into a new market.

Fact 2: frontier use is concentrated in task families common in digitally tradable services. Mapped 1P API traffic is concentrated in a small set of O*NET tasks. Direct calculation from the release files puts the record-weighted top-ten share at around 40 percent in February 2026. These are task families that overlap with software and IT services, back-office and finance-and-accounting work, and contact-centre and business-process services—the activities that the optimistic view treats as new rungs and the pessimistic view treats as vulnerable ones. In the May 2026 chat aggregate, Computer and Mathematical is likewise the top job category of both Indian and Philippine usage (26.2 and 26.7 percent of own usage, against a 23.8 percent world baseline), and

the software-development request topic runs at 16.2 and 16.6 percent against 11.5 worldwide.

Three distinct objects are easily compressed here, and the model keeps them apart. *Task exposure* is what these data can help locate. *Sectoral tradability* is not observed at all: nothing in the AEI says whether an activity is exported. And the *ladder weight* ψ_s —how much a sector’s work builds capability—has to come from historical evidence on learning, and is a separate dimension from exposure by construction. Concretely, the AEI locates candidate values of the task-level intensity π_k^{AI} in the aggregation (21) that builds sectoral exposure a_s ; it does not establish that the corresponding activity is exported, that it has a high ladder weight, or that its labour-demand effect is positive or negative—which by Proposition 1 turns on a world demand elasticity that no usage dataset contains.

The model’s manufacturing channel M includes AI embodied in robotics and process control, and the Anthropic data are almost entirely about language-model interaction. They are therefore close to silent on a_M . Instead, I draw from the robotics and automation literature, e.g. [Acemoglu and Restrepo \(2018\)](#).

Fact 3: lower observed usage intensity is associated with a higher automation share.

The automation share of conversations falls by 2.10 percentage points per log point of the AUI : countries in the bottom AUI quartile average 52.6 percent automation-style use against 46.7 in the top quartile—Kenya at 54.6 and India at 51.9, against the United Kingdom at 44.8 and Germany at 45.2. Decomposing the gradient against each country’s published job-category shares assigns -0.05 (0.09) to composition and -2.05 (0.18) to the within-category margin (Figure 2 and Table 2). Standardizing each country’s delegation to the world’s within every occupation, holding its own occupation mix fixed, produces a predicted gradient of $+0.00$ (0.27), while the within-occupation residual carries -2.47 (0.41) of the observed -2.47 —composition accounts for none of the relationship and the within-occupation margin for all of it (Table 3). Conditional on broad occupational composition, in other words, lower-AUI countries exhibit higher automation shares.

This motivates the model’s treatment of the delegation mode as a margin distinct from implementation, and behind the two-edged character of absorption in Section 3.5: the same implementation that defends output can be the configuration in which work teaches least. The March 2026 report finds that the education level required to understand user prompts rises by almost a school-year per additional year of Claude use, tapering after roughly a year, alongside higher task-success rates for experienced users ([Anthropic, 2026a](#)). That is consistent with users learning the tool, with selection into sustained use, and with task upgrading; it does not sign the learning content of the tasks delegated at work.

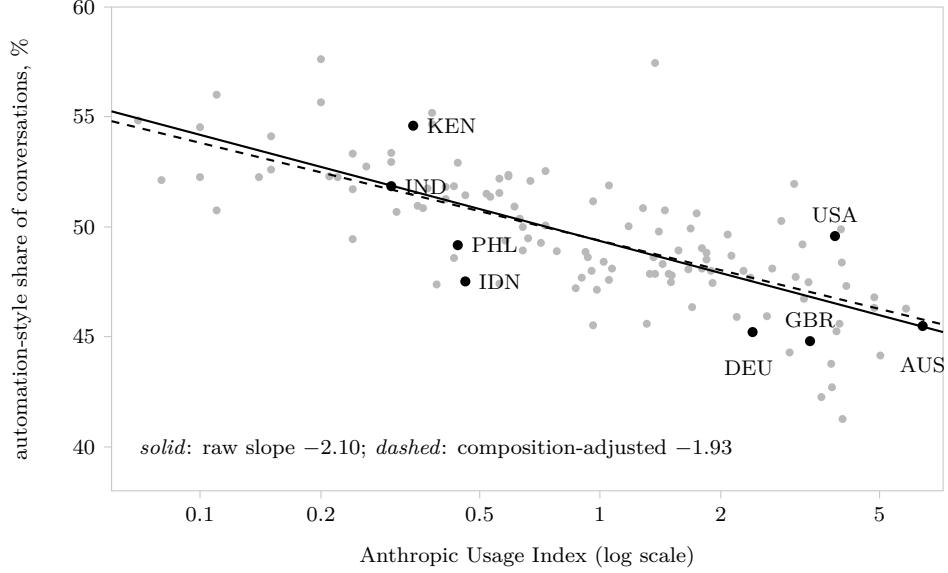


Figure 2: Delegation mode against usage intensity: automation-style share of conversations against the Anthropogenic Usage Index, May 2026 aggregate. Source: [Anthropic \(2026c\)](#).

3 Model

Time is continuous. At each instant the economy is characterized by the exogenous AI frontier $F \in \mathbb{R}_+$ and two predetermined domestic stocks: productive capability $H \in \mathbb{R}_+$, which governs learning, and AI readiness $D \in \mathbb{R}_+$, which governs implementation. The labour endowment is fixed at one and labour is freely mobile across sectors.

Carrying two stocks rather than one follows directly from what the readiness evidence of Section 2 looks like. Productive capability accumulates through learning by doing and is by construction slow-moving; observed AI readiness is not, since it moves when a product is launched, localized, repriced, or made available under a new licence. Merging them would make high- H economies mechanically both better learners and better adopters, and would force a slow-moving state to carry a fast-moving observable.

3.1 Sectors

There are three sectors:

- M : manufacturing tradables,
- P : progressive tradable services,
- C : consumer non-tradable services.

The first two are capability-building tradables; the third is the local, labour-absorbing service sector. In the data, P should be read narrowly—ICT, finance, professional and business services,

tradable back-office and digitally delivered services—and C broadly, as the local (and in much of the developing world largely informal) urban service economy: retail, hospitality, transport, personal services. The model abstracts from agriculture to focus on the post-agricultural development margin that is central to the AI debate. Throughout, S denotes the set of tradable sectors; the baseline takes $S = \{M, P\}$, but the main theorem holds verbatim for any finite nonempty S .

The two-sector baseline should be read as an expository device rather than as the recommended empirical partition. Pooling software engineering, high-end professional services, accounting and finance-and-accounting back offices, routine data entry, and contact-centre work into a single P averages across precisely the heterogeneity the covariance term is built to detect: these activities need not share a learning weight ψ_s , an AI intensity a_s , a world demand elasticity ζ_s , or a residual demand elasticity μ_s , and the optimistic and pessimistic narratives about AI and services are, in part, narratives about different halves of this set. The minimal useful refinement is

$$S = \{M, P^a, P^r\}, \quad (1)$$

with P^a advanced tradable services—software and IT services, engineering, high-end professional and financial services—and P^r routine tradable services—business-process outsourcing, transaction processing, contact-centre and standardized back-office work. Nothing below changes: Theorem 1 is stated for arbitrary finite S , and the covariance term simply runs over three sectors instead of two. What changes is the empirical content, because the pessimistic case for a dissolving service ladder is sharpest exactly where ψ_{P^r} is low and a_{P^r} high, and that configuration is invisible inside a pooled P . The two-sector notation is retained in the text for readability, and (1) is the partition the empirical implementation of Section 9 should use.

3.2 Households

At a point in time the representative household consumes an imported tradable composite T and local non-tradable services C . Flow utility is the log of a CES aggregator,

$$u(T, C) = \left[\omega T^{\frac{\sigma-1}{\sigma}} + (1-\omega)C^{\frac{\sigma-1}{\sigma}} \right]^{\frac{\sigma}{\sigma-1}}, \quad 0 < \omega < 1, \quad 0 < \sigma < 1, \quad (2)$$

and intertemporal preferences, used only for the welfare analysis of Section 6, are $\int_0^\infty e^{-\rho t} \ln u(T_t, C_t) dt$ with $\rho > 0$. The restriction $0 < \sigma < 1$ is the gross-complements case; it is exactly the force that delivers Baumol reallocation toward local services when their relative price rises.

The tradable composite is the numeraire. Let q denote the price of C and w the wage; with zero profits and a unit labour endowment, household income equals w , and households are hand-to-mouth: the period budget constraint is

$$T + qC = w, \quad (3)$$

so trade is balanced instant by instant. The CES demand system implies an expenditure share on

non-tradables of

$$s_C(q) \equiv \frac{qC}{w} = \frac{(1-\omega)^\sigma q^{1-\sigma}}{\omega^\sigma + (1-\omega)^\sigma q^{1-\sigma}}. \quad (4)$$

Because $0 < \sigma < 1$, the expenditure share on C is increasing in its relative price:

$$\frac{ds_C(q)}{d \ln q} = (1-\sigma)s_C(q)(1-s_C(q)) > 0. \quad (5)$$

Equation (5) is the Baumol channel in reduced form; only this property of demand is used in the baseline, and Appendix B replaces (4) with a general—possibly non-homothetic—share function.

3.3 Technology and world demand

Sector C is non-tradable with linear technology,

$$Y_C = A_C(F)L_C, \quad a_C \equiv \partial_F \ln A_C(F) \geq 0, \quad (6)$$

where $A_C : \mathbb{R}_+ \rightarrow (0, \infty)$ is continuously differentiable. Local services are thus allowed to absorb AI too. The *unexposed-services benchmark* $a_C = 0$ is the special case in which frontier AI does nothing to local non-tradable productivity. Under competition,

$$q = \frac{w}{A_C}. \quad (7)$$

Because value added equals the wage bill, the employment share of the non-tradable sector equals the expenditure share on that sector:

$$L_C = s_C(q). \quad (8)$$

Tradable sectors $s \in S$ produce with linear technologies,

$$Y_s = A_s(H, F)L_s, \quad (9)$$

with $A_s : \mathbb{R}_+^2 \rightarrow (0, \infty)$ continuously differentiable and nondecreasing in H , and sell on world markets. The home country faces residual isoelastic demand in each tradable sector:

$$p_s = \Xi_s(F)Y_s^{-1/\mu}, \quad \mu > 1, \quad (10)$$

where $\Xi_s : \mathbb{R}_+ \rightarrow (0, \infty)$ is a residual-demand shifter; a decline in Ξ_s captures tougher foreign competition. Section 3.5 derives (10) and the behavior of $\Xi_s(F)$ from world demand primitives. Under perfect competition, unit prices equal unit labour costs, $p_s = w/A_s(H, F)$, so equilibrium output and labour demand in sector s are

$$Y_s = \left(\frac{\Xi_s(F)A_s(H, F)}{w} \right)^\mu, \quad L_s = \Xi_s(F)^\mu A_s(H, F)^{\mu-1} w^{-\mu}. \quad (11)$$

Define

$$\Omega_s(H, F) \equiv \Xi_s(F)^\mu A_s(H, F)^{\mu-1}, \quad \Omega(H, F) \equiv \sum_{s \in S} \Omega_s(H, F), \quad (12)$$

so that

$$L_s = \Omega_s(H, F)w^{-\mu}, \quad L_T \equiv \sum_{s \in S} L_s = \Omega(H, F)w^{-\mu}. \quad (13)$$

3.4 AI: absorption and foreign competition in reduced form

The model treats frontier AI as a small shift in F . Domestic absorption is sector-specific and depends on both domestic stocks.

Assumption 1 (Reduced-form absorption and competition). *For each tradable sector $s \in S$,*

$$\partial_F \ln A_s = \alpha_s(H, D), \quad (14)$$

where $\alpha_s : \mathbb{R}_+^2 \rightarrow \mathbb{R}_+$ is continuously differentiable, nondecreasing in each argument and strictly increasing in D , and

$$\partial_F \ln \Xi_s(F) = -\nu_s, \quad \nu_s > 0. \quad (15)$$

In addition, at each H ,

$$(\mu - 1)\alpha_s(H, 0) < \mu\nu_s < (\mu - 1) \lim_{D \rightarrow \infty} \alpha_s(H, D). \quad (16)$$

Define the *fixed-wage labour-demand effect* of a marginal frontier advance in tradable sector s by

$$\vartheta_s(H, D) \equiv (\mu - 1)\alpha_s(H, D) - \mu\nu_s, \quad (17)$$

so that, using (12),

$$\partial_F \ln \Omega_s = \vartheta_s(H, D). \quad (18)$$

That is, $\vartheta_s(H, D)$ is the log change in sector- s labour demand induced by a frontier AI increment at a given wage. ϑ_s is *not* a competitiveness effect. It bundles the response of home unit costs, home market share, and the size of the world market for the sector, and Proposition 2 below separates those components—a separation with real consequences, because domestic absorption turns out to govern only one of them.

Remark 1 (Notation: two states, one period economy). *Both H and D are predetermined within the period, and the static block treats them symmetrically: they enter only through the values α_s , and hence through Ω_s . To keep the notation light, the readiness argument is displayed only where it does work—in (14)–(17), in the microfoundation, in the threshold results, and in the theorem—and is suppressed elsewhere. Where a one-dimensional state is genuinely needed—the transition dynamics of Section 7, the fold analysis, and the convergence speed κ —we impose the diagonal convention*

$$D = d(H), \quad d : \mathbb{R}_+ \rightarrow \mathbb{R}_+ \text{ continuous and nondecreasing}, \quad (19)$$

of which $D \equiv H$ is the leading case, and write $\alpha_s(H) \equiv \alpha_s(H, d(H))$, $\vartheta_s(H) \equiv \vartheta_s(H, d(H))$. The convention is a modelling choice made for tractability at one point in the argument.

3.5 Where thresholds come from: a task-level microfoundation

Consider the following primitives, in the spirit of the task framework of [Acemoglu and Restrepo \(2018\)](#) and the absorption-lag evidence of [Comin and Mestieri \(2018\)](#). Production in tradable sector s consists of a continuum of tasks, a fraction of which can be delegated to frontier AI at each frontier level; delegating a task lowers unit costs, and the flow of such opportunities arrives at sector-specific intensity $a_s > 0$ per unit of F . A country with readiness D successfully implements the share $g(D)$ of these opportunities, where $g : \mathbb{R}_+ \rightarrow [0, 1]$ is continuous and strictly increasing with $\lim_{D \rightarrow \infty} g(D) = 1$; frontier economies implement them all. Implementation is placed in readiness rather than capability deliberately: what determines whether an available task is actually delegated is infrastructure, compute and model access, licensing, language coverage, and complementary technical skills, none of which is the learning-by-doing stock that (26) accumulates. Home productivity is therefore

$$\ln A_s = \ln \bar{A}_s(H) + a_s g(D) F, \quad \text{so} \quad \alpha_s(H, D) = a_s g(D). \quad (20)$$

The two stocks accordingly do different jobs in the microfoundation: H sets the pre-AI productivity level $\bar{A}_s(H)$ and, through (26), the ladder weights; D sets the rate at which the frontier is converted into domestic productivity. The sectoral intensity a_s is the object the task composition of frontier use speaks to. If sector s 's pre-AI production uses task family k with weight ω_{sk} and π_k^{AI} measures the intensity with which frontier systems are used on family k , then a natural empirical construction is

$$a_s \propto \sum_k \omega_{sk} \pi_k^{\text{AI}}, \quad (21)$$

which is the bridge between a task-level usage measure and a sector-level structural parameter. Fact 2 of Section 2 bears on π_k^{AI} and on nothing else in (21): the pre-AI task weights ω_{sk} come from occupational employment or input data, and the proportionality constant is a normalization of the frontier scale F . Section 9.3 states the exporter-level version of (21) and what it would take to compute it. On world markets, home output is one variety in a sector- s CES aggregate dominated by frontier suppliers, whose price index $\bar{P}_s(F)$ falls with the frontier at the full absorption rate, $\partial_F \ln \bar{P}_s = -a_s$. World demand for the sector has price elasticity $\zeta_s \geq 0$: total world purchases are $\mathcal{D}_s(F) = \bar{\mathcal{D}}_s \bar{P}_s(F)^{-\zeta_s}$. Home residual demand is then $Y_s = \mathcal{D}_s(F) (p_s / \bar{P}_s(F))^{-\mu}$, which is exactly (10) with

$$\Xi_s(F) = \bar{\mathcal{D}}_s^{1/\mu} \bar{P}_s(F)^{\frac{\mu - \zeta_s}{\mu}}, \quad \nu_s = a_s \frac{\mu - \zeta_s}{\mu}. \quad (22)$$

Foreign competition is thus not a free parameter: it is frontier absorption a_s filtered through the gap between how easily buyers substitute across suppliers (μ) and how much cheaper provision expands the world market (ζ_s).

The microfoundation is strictly wider than the reduced form it is meant to justify: Assumption 1 imposes $\nu_s > 0$, which by (22) is exactly $\zeta_s < \mu$. For $\zeta_s \geq \mu$ the microfoundation remains perfectly coherent—the expansion of the world market then outweighs the decline in home export prices, so $\nu_s \leq 0$ and the residual-demand shifter Ξ_s weakly *rises* with the frontier—but that configuration lies outside the reduced form’s “tougher foreign competition” region. All reduced-form statements below should therefore be read as maintaining $\zeta_s < \mu$; the microfounded pure-expansion regime of Proposition 1(ii) extends beyond it.

Proposition 1 (Emergent threshold structure). *Under the microfoundation (20)–(22),*

$$\vartheta_s(H, D) = a_s \left[(\mu - 1)g(D) - (\mu - \zeta_s) \right], \quad (23)$$

which does not depend on H , and three regimes obtain:

- (i) Pure displacement ($\zeta_s \leq 1$): $\vartheta_s < 0$ at every readiness level. No level of absorptive capacity makes frontier AI raise sector- s labour demand at a fixed wage.
- (ii) Pure expansion ($\zeta_s \geq \mu - (\mu - 1)g(0)$): $\vartheta_s > 0$ for every $D > 0$, and $\vartheta_s|_{D=0} \geq 0$ with equality if and only if $\zeta_s = \mu - (\mu - 1)g(0)$. labour demand is thus weakly positive at zero readiness and strictly positive thereafter.
- (iii) Interior threshold ($1 < \zeta_s < \mu - (\mu - 1)g(0)$): there is a unique $\underline{D}_s > 0$ with $\vartheta_s|_{D=\underline{D}_s} = 0$, given by

$$g(\underline{D}_s) = \frac{\mu - \zeta_s}{\mu - 1}, \quad (24)$$

and the endpoint condition (16) of Assumption 1 holds. The required implementation share is decreasing in ζ_s and increasing in μ . Under the diagonal convention (19) with d strictly increasing, the corresponding capability threshold is $\underline{H}_s = d^{-1}(\underline{D}_s)$.

Proof. Substituting $\alpha_s = a_s g(D)$ and $\nu_s = a_s(\mu - \zeta_s)/\mu$ into (17) gives (23). Since g is strictly increasing with range $[g(0), 1)$, the function $D \mapsto \vartheta_s$ is strictly increasing with infimum $a_s[(\mu - 1)g(0) - \mu + \zeta_s]$ and supremum $a_s(\zeta_s - 1)$. Case (i): if $\zeta_s \leq 1$ the supremum is nonpositive and is not attained, so $\vartheta_s < 0$ everywhere. Case (ii): if $\zeta_s \geq \mu - (\mu - 1)g(0)$ the infimum $a_s[(\mu - 1)g(0) - \mu + \zeta_s]$, attained at $D = 0$, is nonnegative—zero exactly at the boundary $\zeta_s = \mu - (\mu - 1)g(0)$ —and strict monotonicity gives $\vartheta_s > 0$ for $D > 0$. Case (iii): the infimum is negative and the supremum positive, so by continuity and strict monotonicity there is a unique zero, at (24). Monotonicity of the threshold follows from $\partial_\zeta[(\mu - \zeta)/(\mu - 1)] < 0$ and $\partial_\mu[(\mu - \zeta)/(\mu - 1)] = (\zeta - 1)/(\mu - 1)^2 > 0$ in the interior regime. $\square$

Equation (23) makes the division of labour between the two stocks exact under the microfoundation: the escalator condition is a condition on *readiness*, while what the escalator delivers—capability—is accumulated by H . A country can therefore be AI-ready and capability-poor, clearing every threshold in Proposition 1, defending its market shares, and still accumulating little if the

labour-demand gains land in low- ψ sectors. That two-dimensional alignment is exactly what the covariance term of Theorem 1 measures, and it is invisible in a one-stock model.

The microfoundation also makes precise what ϑ_s does and does not measure. Let $E_s \equiv \bar{P}_s \mathcal{D}_s$ denote world expenditure on sector s and $m_s \equiv p_s Y_s / E_s$ the home revenue share of that expenditure.

Proposition 2 (What ϑ_s is and is not). *Under the microfoundation (20)–(22), a marginal frontier advance at a fixed wage moves, in sector s :*

- (i) home costs relative to frontier suppliers: $\partial_F \ln(p_s / \bar{P}_s) = a_s(1 - g(D)) \geq 0$;
- (ii) home market share: $\partial_F \ln m_s = -a_s(\mu - 1)(1 - g(D)) \leq 0$, independent of ζ_s and equal to zero under full absorption;
- (iii) world expenditure on the sector: $\partial_F \ln E_s = a_s(\zeta_s - 1)$, independent of home absorption;
- (iv) home output: $\partial_F \ln Y_s = a_s[\zeta_s - \mu(1 - g(D))]$;
- (v) home employment and wage bill: $\partial_F \ln L_s = \vartheta_s(H, D) = \partial_F \ln m_s + \partial_F \ln E_s$.

Proof. At fixed w , $p_s = w/A_s$ gives $\partial_F \ln p_s = -a_s g(D)$, while $\partial_F \ln \bar{P}_s = -a_s$; (i) follows. From $E_s = \bar{\mathcal{D}}_s \bar{P}_s^{1-\zeta_s}$, (iii) follows. From $Y_s = \mathcal{D}_s(F)(p_s/\bar{P}_s)^{-\mu}$, $\partial_F \ln Y_s = a_s \zeta_s - \mu a_s(1 - g)$, which is (iv). Revenue is $p_s Y_s = w L_s$, so $\partial_F \ln L_s = \partial_F \ln p_s + \partial_F \ln Y_s = a_s[(\mu - 1)g - (\mu - \zeta_s)] = \vartheta_s$ by (23), and $\partial_F \ln m_s = \partial_F \ln(p_s Y_s) - \partial_F \ln E_s = \vartheta_s - a_s(\zeta_s - 1) = -a_s(\mu - 1)(1 - g)$, giving (ii) and (v). $\square$

Proposition 2 disciplines the interpretation of the three regimes. The fixed-wage labour-demand effect decomposes exactly into a *market-share effect*, which is competitiveness proper—it depends only on the absorption gap $1 - g(D)$, and full absorption neutralizes it for *any* world demand elasticity—and a *world-market-size effect*, which depends only on ζ_s and which no amount of domestic absorption can touch. Regime (i) of Proposition 1 is therefore *not* a statement that absorption is futile. A country with $g(D)$ near one keeps its market share and, at a fixed wage, its output response tends to $\partial_F \ln Y_s|_w \rightarrow a_s \zeta_s \geq 0$ —strictly positive whenever $\zeta_s > 0$ —even when $\zeta_s \leq 1$; what contracts is *world expenditure on the sector*, and with it the home sector’s fixed-wage wage bill, because productivity outruns inelastic world demand. (At the boundary $\zeta_s = 1$ world expenditure is unchanged rather than shrinking, and under full absorption the fixed-wage wage bill merely stops growing.) Absorption defends competitiveness, output, and—if learning attaches to output rather than employment (Appendix D)—possibly capability; what it cannot defend, when $\zeta_s \leq 1$, is employment. The development claims of this paper, which run through the labour-based learning technology (26), are conditional statements about that employment margin, and Section 9 treats the distinction as an estimation requirement rather than a footnote.

With that qualification, Proposition 1 carries three lessons. First, an interior *employment* escalator threshold requires $\zeta_s > 1$: AI-driven cost declines must expand the world market for the sector more than one-for-one—a Jevons condition on world expenditure, item (iii) above. The policy debate about AI-readiness implicitly assumes regime (iii) of Proposition 1; whether that

assumption holds is an empirical question about world demand curves, answerable from trade data alone. Second, a higher μ —home output more substitutable with frontier output, more “commodity-like”—raises the absorption bar (24). Moving up within a sector (differentiation) and absorbing AI are complements. Third, the AI-intensity a_s scales the magnitude of ϑ_s but not its sign: heavily AI-exposed sectors are where the stakes are largest in both directions, which is exactly why exposure measures alone cannot sign the development effect.

How much is adopted vs how it is implemented

The same task structure microfounds the learning-content channel announced in the introduction, but doing so requires a margin the implementation share g does not contain. Whether a task is delegated at all is one decision; whether the worker directs the system and stays in the task, or hands the task over outright, is another. The first is the readiness margin already in the model. The second—call it the *delegation mode*—is what Fact 3 of Section 2 describes, and in a one-margin model there is nowhere to put it.

Let $d_s(H, D, F) \in [0, 1]$ be the share of sector s ’s *implemented* uses that are substitutive rather than complementary: the fraction of delegated tasks the system executes outright rather than under the worker’s direction. Suppose that complementary use teaches at rate $\tau_s \geq 0$ —the tutor, codification, and task-upgrading channel—while substitutive use forgoes the learning content of the delegated task at rate $\eta_s \geq 0$. Then delegation changes the sector’s learning weight at rate

$$\partial_F \ln \psi_s = a_s g(D) [(1 - d_s) \tau_s - d_s \eta_s] \Psi'(F), \quad \tau_s, \eta_s \geq 0, \quad d_s \in [0, 1], \quad (25)$$

where the delegation profile $\Psi : \mathbb{R}_+ \rightarrow \mathbb{R}_+$ is continuously differentiable and bounded with $\Psi' > 0$ and $\Psi(0) = 0$, $\Psi_{\sup} \equiv \sup_{F \geq 0} \Psi(F) < \infty$, and is one and the same function for every $s \in S$. The commonality of Ψ is what makes the frontier factor $\Psi'(F)$ scale all sectors alike, so that it changes neither the sign of the learning-content term nor the relative sectoral contributions; sector-specific profiles are admissible but lose that property.

First, it separates *how much* is implemented from *how* it is implemented: $g(D)$ scales the channel, d_s signs it. The net learning-content rate $(1 - d_s)\tau_s - d_s\eta_s$ is positive below the mode threshold $d_s < \tau_s/(\tau_s + \eta_s)$ and negative above it, so the same sector with the same exposure a_s and the same readiness can be capability-building in one country and capability-eroding in another, purely through the organization of use. A single sectoral constant η_s of free sign is the special case $d_s \equiv 1$ of (25) when $\eta_s \geq 0$, and $d_s \equiv 0$ is the case in which the tutor channel is live. Second, the boundedness that Assumption 2 imposes survives: since $|(1 - d_s)\tau_s - d_s\eta_s| \leq \max\{\tau_s, \eta_s\}$ and $g < 1$,

$$|\ln \psi_s(H, F) - \ln \psi_s(H, 0)| \leq a_s \max\{\tau_s, \eta_s\} \Psi_{\sup},$$

so $\sup \psi_s$ is finite whenever the $F = 0$ profile is, for *any* path of d_s and either sign of the net effect. Only a finite stock of tasks is ever delegated. As before this makes the microfoundation compatible with Assumption 2 rather than deriving it, and the constant-rate case $\Psi(F) = F$ remains

inadmissible, since it makes ψ_s unbounded whenever the net rate is negative. Third, d_s is written as a function of the state and the frontier, not as a parameter, because it is an equilibrium object: how a firm splits a task between worker and model depends on what the model can do, on the price of the worker’s time, and on the complementary capability available to supervise the output. Fact 3 is suggestive of d_s falling with readiness over the observed range. It does not identify that relationship, is measured across countries rather than within a sector, and—because the automation share is a share of *observed* conversations rather than of implemented production tasks—does not pin the level of d_s either.

If AI takes over the learning-rich tasks outright—the apprenticeship layer of junior analysis, drafting, and debugging—then $d_s\eta_s$ dominates and absorption hollows out learning; if AI acts as a tutor, codifies organizational knowledge, or lets junior workers perform senior tasks under supervision, then $(1 - d_s)\tau_s$ dominates and absorption enriches it. Evidence that generative AI produces its largest gains among less experienced workers (Brynjolfsson et al., 2025) shows the second case is live, not hypothetical. Equation (25) has a sharp implication either way: at a given frontier level, the learning-content effect scales with *own* implementation $g(D)$. Where the net rate is negative, the same absorption that defends market share and, at a fixed wage, employment (Proposition 2) simultaneously erodes what the remaining jobs teach—absorption is two-edged, and the trade-off is invisible to any policy framework that scores AI-readiness alone. That is also why readiness policy and ladder policy are not substitutes: raising D moves the economy up g , but only the composition of use determines what the resulting work teaches.

Equation (20) lets the mode of use affect learning but not productivity: the productivity gain per implemented task is the same whether the task is executed outright or performed under direction. If automation and augmentation differ in their productivity or labour-displacement content—plausible, and an open empirical question—then d_s belongs in a_s and hence in α_s and ϑ_s as well, with a_s replaced by $a_s(d_s)$ throughout. Every result below goes through with that substitution, since the static block uses only the values of α_s ; what changes is that the mode margin then moves the covariance term as well as the learning-content term, and the two need not move together. The baseline keeps them separate because it is the cleaner statement of what the mode margin does that the exposure margin cannot.

The baseline analysis proceeds under the reduced form (Assumption 1), which nests regime (iii) and does not require the specific functional forms of this subsection; the learning-content channel enters through the general dependence $\psi_j(H, F)$ introduced next. Where (25) is used, it is not unrestricted: it drifts the log ranking gap between tradable sectors, so an ordering such as $\psi_P \geq \psi_M$, where it is invoked, additionally requires

$$\ln \psi_P(H, 0) - \ln \psi_M(H, 0) \geq g(D) \Psi_{\sup} \sup\{a_M[(1 - d_M)\tau_M - d_M\eta_M] - a_P[(1 - d_P)\tau_P - d_P\eta_P]\},$$

for which $\ln \psi_P(H, 0) - \ln \psi_M(H, 0) \geq g(D) \Psi_{\sup} (a_P\eta_P + a_M\tau_M)$ is sufficient—an inequality between the initial ladder gap and the total delegable stock, which fails once $\Psi_{\sup}$ is large enough. This is one of the reasons the ordering is not maintained as a global assumption below.

3.6 Capability accumulation

Capability evolves according to the sectoral allocation of labour and the learning content of each sector's work:

$$\dot{H} = \sum_{s \in S} \psi_s(H, F) L_s + \psi_C(H, F) L_C - \delta H, \quad \delta > 0. \quad (26)$$

Assumption 2 (State-dependent ladder weights). *Each $\psi_j : \mathbb{R}_+^2 \rightarrow \mathbb{R}_+$, $j \in S \cup \{C\}$, is continuous, bounded, and continuously differentiable in F , and for all (H, F) and every tradable sector $s \in S$,*

$$\psi_s(H, F) > \psi_C(H, F) \geq 0. \quad (27)$$

The learning-content responses $\partial_F \psi_j(H, F)$ are unrestricted in sign, and the ladder weights are not ordered among themselves. For the results that differentiate the capability drift field in H —the convergence speed κ of Proposition 8, the branch structure of Section 7, and the co-state equation behind Proposition 11—each ψ_j is in addition continuously differentiable in H , with $\partial_H \psi_j$ bounded on the state space; the static results of Sections 4 and 5 do not use this.

What is maintained is only that tradable activities build capability more than local consumer services, which is what the Baumol term needs ($\bar{\psi} - \psi_C > 0$). The ordering *within* the tradable block is deliberately left open: the paper does not maintain $\psi_P \geq \psi_M$ —progressive tradable services at least as capability-building as manufacturing—for three reasons. It is not needed: Theorem 1 holds for any finite S with arbitrary ψ_s , and the covariance term (45) is signed by whether the two rankings agree, not by which sector sits on top. It is not implied by the evidence: the case for P being the higher rung is an argument about software and high-end professional services, and P as defined below also contains routine back-office and contact-centre work whose learning content is an open question. And maintaining it while the learning-content channel is live requires the awkward side condition of Section 3.5, an inequality between the initial ladder gap and the total delegable stock that has no independent justification. Where the ordering is genuinely useful—as a reading of the optimistic “new rungs” narrative—it is invoked locally and flagged. Constant ladder weights are the special case $\psi_j(H, F) \equiv \psi_j$, and the frontier-independent case $\psi_j(H, F) = \psi_j(H)$ shuts down the learning-content channel of Section 3.5. Boundedness, together with $\psi_j \geq 0$, $\delta > 0$ and labour-market clearing (28), implies that the interval $\mathcal{K} \equiv [0, \bar{\psi}_{\text{sup}}/\delta]$, with $\bar{\psi}_{\text{sup}} \equiv \sup_{j, H, F} \psi_j(H, F)$, is *forward* invariant for (26): clearing is what bounds $\sum_j \psi_j L_j$ above by $\bar{\psi}_{\text{sup}}$, and invariance is one-sided, since $\dot{H} > 0$ at $H = 0$ carries the backward trajectory out of $\mathcal{K}$ immediately. All dynamic statements are on $\mathcal{K}$. Workers are paid their marginal product and no agent internalizes (26): capability accumulation is a pure externality, a fact that Section 8 exploits.

Two aspects of (26) are substantive economic assumptions, not normalizations, and both matter for what follows. First, the *basis* of learning is worker-years: capability accrues in proportion to employment, so a sector that produces more output with fewer workers contributes less learning. This is exactly the margin AI moves—AI is a technology for producing more output, quality, and export intensity per worker—so the choice of basis is consequential. If learning instead attaches

to output, exports, or buyer relationships, as the experimental evidence on exporting suggests it may (Atkin et al., 2017), the Baumol mechanism weakens and can reverse; Appendix D derives the multiplier under the output-based alternative and states the empirical moment that discriminates between the two. Second, the ladder weights attach to H and to F but not to readiness D : what a job teaches depends on the accumulated capability of the economy around it and on how much the frontier has changed the job, not on how much compute the country has bought. That is a modelling judgement rather than a theorem, and it is what gives the two stocks their division of labour— D governs implementation, H governs learning. Appendix E gives the accumulation law for D and the planar dynamics that the diagonal convention (19) stands in for.

3.7 Discussion of maintained assumptions

Remark 2 (Hand-to-mouth and balanced trade). *Households cannot borrow internationally, so trade balances instant by instant and the static equilibrium is a sequence of period equilibria linked only through H . Intertemporal trade would let a country smooth through a transitory loss of competitiveness but would not change the composition logic, which operates through relative labour demands.*

Remark 3 (Informality). *The C sector is best read as including the informal urban service economy. The model’s “Baumol pull” into C is then the familiar observation that displaced or priced-out labour in developing countries is absorbed into low-productivity informal services rather than unemployment. The common wage and frictionless mobility sit uneasily with that reading: informal absorption typically occurs at lower earnings, and the workers absorbed are rarely those who could staff progressive tradable services. Appendix F introduces the minimal heterogeneity—two labour types, with only skilled workers able to enter P —and shows what survives: the decomposition holds on the unskilled margin, but when the skill constraint binds, AI gains in progressive services arrive as skilled rents that pull unskilled labour from manufacturing into local services, and the “invisibility” results of Section 6 weaken in exactly the direction one would expect—the trap acquires a contemporaneous signature in the skill premium and in unskilled earnings, which can fall even as aggregate income rises.*

Remark 4 (Tradability and learning are different dimensions). *The baseline ranking (27) makes both tradable sectors more capability-building than local services, but tradability and learning intensity are conceptually orthogonal: education, healthcare, and sophisticated local professional services are high-learning non-tradables, while some standardized tradable tasks teach little. The framework accommodates this. The theorem below holds for any finite set of tradable sectors with arbitrary ψ_s , and the learning content of the non-tradable block enters only through $\bar{\psi} - \psi_C$: a high- ψ_C local sector shrinks the Baumol penalty toward zero without changing the covariance term. Empirically, C should be split so that high-learning non-tradables are not forced to carry the low ψ_C of informal retail; the model’s prediction is that the Baumol reallocation is damaging only insofar as it lands in genuinely low-learning activities.*

Remark 5 (AI inside local consumer services). *Because the exposure of local services is a first-order determinant of the Baumol term, a_C is carried in the main results (Proposition 5, Theorem 1, Definition 1) rather than treated as an extension; the benchmark $a_C = 0$ is displayed wherever it is invoked. Two channels remain outside even that treatment. AI could raise the quality of local services or shift demand toward them, expanding C employment through a taste or quality margin the model does not carry; and it could raise ψ_C , the learning content of local-service work, which enters the multiplier through $L_C \partial_F \psi_C$ and is the one term in Theorem 1 that a purely tradable-sector empirical design will never see. Both work against reading the Baumol reallocation as automatically damaging.*

Remark 6 (AI as an imported, paid-for input). *Frontier models, compute, and cloud services are largely supplied by firms in advanced economies. Domestic absorption is then not a free productivity shifter: gross productivity gains exceed the gains in domestic value added by the rents paid to foreign AI suppliers. In the model's terms, α_s should be read as the productivity gain net of imported-input payments at market prices, which is the right object for wages and labour demand; but the distinction resurfaces in welfare and policy, where GDP-style gains overstate GNI-style gains, and where taxing or negotiating foreign AI rents becomes an instrument distinct from the ladder subsidies of Section 8. The reduced form absorbs the level; it cannot address the division of surplus, which we flag as a first-order question for the quantitative work.*

Remark 7 (Capital). *If AI adoption is embodied in capital, absorption becomes $\alpha_s(H, K)$ with $\partial_K \alpha_s > 0$, creating an H – K complementarity that typically sharpens thresholds, while cheap imported AI capital relaxes them. The labour-only model is a lower-dimensional core, not a claim that capital is irrelevant. (See Appendix G.)*

4 Static equilibrium

Given (H, F) , a static equilibrium is a wage $w > 0$ and labour allocation (L_M, L_P, L_C) satisfying (8), (13), and labour-market clearing:

$$\sum_{s \in S} L_s + L_C = 1. \quad (28)$$

Using (8), (13), and (7), the equilibrium wage is pinned down by

$$\Phi(w; H, F) \equiv \Omega(H, F) w^{-\mu} + s_C \left(\frac{w}{A_C} \right) - 1 = 0. \quad (29)$$

Proposition 3 (Existence and uniqueness of the static equilibrium). *For every $(H, F) \in \mathbb{R}_+^2$, there exists a unique static equilibrium, and it is interior: $L_C \in (0, 1)$ and $L_T \in (0, 1)$.*

Proof. Fix (H, F) and suppress arguments. By (9)–(10) and (12), $0 < \Omega(H, F) < \infty$ at every $(H, F) \in \mathbb{R}_+^2$, S being finite and nonempty. Continuity of Φ is immediate. As $w \downarrow 0$, the first term in

(29) tends to $+\infty$ while $s_C(w/A_C) \rightarrow 0$, so $\Phi(w) \rightarrow +\infty$. As $w \uparrow \infty$, with $q = w/A_C$ and $0 < \sigma < 1$,

$$1 - s_C(q) = \frac{\omega^\sigma}{\omega^\sigma + (1 - \omega)^\sigma q^{1-\sigma}} \sim \left(\frac{\omega}{1 - \omega}\right)^\sigma q^{-(1-\sigma)},$$

and $\Phi(w) = \Omega w^{-\mu} - (1 - s_C(w/A_C))$. Because $\mu > 1 > 1 - \sigma$, the positive term vanishes faster, so $\Phi(w) < 0$ for all sufficiently large w . For uniqueness, the equilibrium condition is equivalent to

$$R(w) \equiv w^\mu \left[1 - s_C\left(\frac{w}{A_C}\right) \right] = \Omega. \quad (30)$$

R is continuous with $R(0^+) = 0$ and $R(\infty) = \infty$ (since $\mu > 1 - \sigma$), and

$$\frac{d \ln R(w)}{d \ln w} = \mu - (1 - \sigma)s_C(w/A_C) \geq \mu - (1 - \sigma) > 0, \quad (31)$$

so R is strictly increasing. Since $\Omega > 0$, there is a unique wage solving (30); (8) and (13) then pin down the allocation. Interiority is immediate from that wage: $0 < w^* < \infty$ and $A_C > 0$ give $q = w^*/A_C \in (0, \infty)$, so $0 < \sigma < 1$ and $0 < \omega < 1$ in (4) give $L_C = s_C(q) \in (0, 1)$ and hence $L_T = 1 - L_C \in (0, 1)$. $\square$

The comparative statics that follow use exactly one scalar statistic of the demand block: the total wage-sensitivity of the local-service employment share,

$$\chi_C(H, F) \equiv \frac{dL_C}{d \ln w}, \quad (32)$$

where the derivative is taken along the equilibrium relations at fixed (H, F) and includes both the relative-price response (since $q = w/A_C$ moves one-for-one with the wage) and, in the general—possibly non-homothetic—demand systems of Appendix B, the income response. Under baseline CES, (5) gives

$$\chi_C = (1 - \sigma)L_C L_T > 0. \quad (33)$$

The results below are stated in terms of χ_C , so the CES demand system is a leading example rather than a load-bearing assumption; what the Baumol mechanism requires is only that a higher wage—through relative prices, income, or both—expands the local-service employment share.

Assumption 3 (Regular demand response). *At every state $(H, F) \in \mathbb{R}_+^2$, $\chi_C > 0$ and $\mu L_T - \chi_C > 0$.*

Assumption 3 holds automatically under CES: by Proposition 3 the equilibrium is interior, so $\chi_C = (1 - \sigma)L_C L_T > 0$, and $(1 - \sigma)L_C < 1 < \mu$ gives $\chi_C < \mu L_T$. Of the two parts, only the non-degeneracy $\mu L_T - \chi_C \neq 0$ is used by the displayed identities that follow; the sign restriction $\chi_C > 0$ is what delivers their directional content and $\mu \Lambda > 1$. Define the object that recurs throughout:

$$\Lambda(H, F) \equiv \frac{L_T}{\mu L_T - \chi_C} \stackrel{\text{CES}}{=} \frac{1}{\mu - (1 - \sigma)L_C(H, F)} \in \left(\frac{1}{\mu}, \frac{1}{\mu - (1 - \sigma)}\right), \quad (34)$$

the general-equilibrium pass-through from log labour-demand shifts to log wages. The wage map is differentiable: by (4) and (31), R is C^1 on $(0, \infty)$ with $R' > 0$, and by the endpoint limits $R(0^+) = 0$, $R(\infty) = \infty$ established in the proof above together with the intermediate value theorem, $R((0, \infty)) = (0, \infty)$; hence R is a C^1 diffeomorphism, so the equilibrium wage is $w^* = R^{-1} \circ \Omega$ and is continuously differentiable in any argument in which Ω is. (In the general—possibly non-homothetic—demand systems of Appendix B the same conclusion follows from the implicit function theorem applied to (29), provided s_C is continuously differentiable; the regularity condition there is $\partial_{\ln w} \Phi = -(\mu L_T - \chi_C) \neq 0$, which is implied by the second part of Assumption 3.) Differentiating labour-market clearing $\Omega w^{-\mu} = L_T = 1 - L_C$ with (32) gives, for any shifter z that moves Ω while leaving A_C fixed,

$$\partial_z \ln w = \Lambda \partial_z \ln \Omega, \quad (35)$$

which under CES coincides with total differentiation of (30). This covers $z = H$ and $z = D$ directly. It covers $z = F$ only under the unexposed-services benchmark $a_C = 0$: a frontier advance also shifts A_C , and Proposition 5 carries the extra term.

5 AI and the direction of structural change

5.1 A first threshold: sectoral labour demand at a fixed wage

Proposition 4 (Fixed-wage employment threshold). *For each tradable sector $s \in S$,*

$$\partial_F \ln L_s|_w = \vartheta_s(H, D). \quad (36)$$

Under Assumption 1 there is, at each capability level H , a unique readiness threshold $\underline{D}_s(H) > 0$ such that $\vartheta_s(H, \underline{D}_s(H)) = 0$, with $\partial_F \ln L_s|_w < 0$ for $D < \underline{D}_s(H)$ and > 0 for $D > \underline{D}_s(H)$. Under the diagonal convention (19) with d strictly increasing, the threshold is a capability level $\underline{H}_s$ solving $\vartheta_s(\underline{H}_s, d(\underline{H}_s)) = 0$, and is unique whenever $H \mapsto \vartheta_s(H, d(H))$ is strictly increasing.

Proof. Holding w fixed in (13) gives $\partial_F \ln L_s|_w = \partial_F \ln \Omega_s = \vartheta_s(H, D)$ by (18). By Assumption 1, $D \mapsto \vartheta_s(H, D)$ is continuous and strictly increasing, negative at $D = 0$ and positive for large D by (16); the intermediate value theorem gives a unique zero. Under (19) the composite $H \mapsto \vartheta_s(H, d(H))$ inherits continuity and the endpoint signs, and strict monotonicity delivers uniqueness. $\square$

Proposition 4 is the cleanest statement of the employment-escalator condition: frontier AI raises sector- s labour demand at a given wage only if domestic absorption is strong enough to offset the induced increase in foreign competition. Under the microfoundation the condition is one on readiness alone, by (23). By Proposition 1, the existence of such a threshold presupposes sufficiently elastic world demand; and by Proposition 2, it is a threshold for employment, not for market share, which absorption defends at every level of ζ_s .

The threshold signs *fixed-wage* labour demand, and the wage is not fixed in equilibrium. Proposition 5 shows that the observed response subtracts a term $\mu \Lambda \bar{\vartheta}$ common to all tradable sectors,

so observed sector- s employment rises if and only if $\vartheta_s > \mu\Lambda\Delta + \mu a_C$, a strictly higher bar than $\vartheta_s > 0$ whenever the average effect $\bar{\vartheta}$ is positive and a strictly lower one whenever it is negative. Sectors with $\vartheta_s > 0$ can shed workers, and—if every tradable sector is hit negatively—sectors with $\vartheta_s < 0$ can gain them, purely through the induced wage. Sections 9.4 and 10 state the predictions in the form the distinction requires.

5.2 General-equilibrium Baumol reallocation

At a given state (H, F) , define within-tradables employment weights and averages

$$\xi_s \equiv \frac{L_s}{L_T} = \frac{\Omega_s}{\Omega}, \quad \bar{\vartheta} \equiv \sum_{s \in S} \xi_s \vartheta_s(H, D), \quad \bar{\psi} \equiv \sum_{s \in S} \xi_s \psi_s(H, F), \quad (37)$$

suppressing arguments (H, F) when no confusion can arise.

Proposition 5 (Wage response, non-tradable expansion, and tradable employment). *Under Assumption 3, and writing $\Delta \equiv \bar{\vartheta} - \mu a_C$ for the relative AI shock to tradables and local services, a marginal increase in the AI frontier changes equilibrium objects according to*

$$\partial_F \ln w = \Lambda \bar{\vartheta} - (\mu\Lambda - 1) a_C = \Lambda \Delta + a_C, \quad (38)$$

$$\partial_F L_C = \chi_C \Lambda \Delta, \quad (39)$$

$$\partial_F \ln L_s = \vartheta_s - \mu\Lambda \Delta - \mu a_C, \quad s \in S. \quad (40)$$

In particular, non-tradable services expand if and only if $\bar{\vartheta} > \mu a_C$, and the wage rises if and only if $\bar{\vartheta} > \chi_C a_C / L_T$, a strictly weaker requirement. A tradable sector may satisfy $\vartheta_s > 0$ and still have $\partial_F L_s < 0$ in equilibrium. Under the unexposed-services benchmark $a_C = 0$ these reduce to $\partial_F \ln w = \Lambda \bar{\vartheta}$, $\partial_F L_C = \chi_C \Lambda \bar{\vartheta}$, and $\partial_F \ln L_s = \vartheta_s - \mu\Lambda \bar{\vartheta}$.

Proof. Since $\partial_F \ln \Omega = \sum_s \xi_s \partial_F \ln \Omega_s = \bar{\vartheta}$ and $\partial_F \ln q = \partial_F \ln w - a_C$ by (7), the definition (32) gives $\partial_F L_C = \chi_C (\partial_F \ln w - a_C)$ under homothetic demand, where the price and income responses enter with the same weight (Appendix B separates them for the non-homothetic case). Differentiating (13) sector by sector gives $\partial_F \ln L_s = \vartheta_s - \mu \partial_F \ln w$, hence $\partial_F L_T = L_T (\bar{\vartheta} - \mu \partial_F \ln w)$. labour-market clearing $\partial_F L_T + \partial_F L_C = 0$ then yields

$$\partial_F \ln w = \frac{L_T \bar{\vartheta} - \chi_C a_C}{\mu L_T - \chi_C} = \Lambda \bar{\vartheta} - (\mu\Lambda - 1) a_C,$$

using $\mu\Lambda - 1 = \chi_C / (\mu L_T - \chi_C)$ from (34), which is (38); substituting back gives $\partial_F L_C = \chi_C (\Lambda \bar{\vartheta} - \mu \Lambda a_C) = \chi_C \Lambda \Delta$ and (40). The two sign conditions follow from $\Lambda > 0$, $\chi_C > 0$, and $(\mu\Lambda - 1)/\Lambda = \chi_C / L_T$. (Consistency: $\partial_F L_T = -\chi_C \Lambda \Delta = -\partial_F L_C$.) $\square$

The term $\mu\Lambda\Delta$ in (40) is the general-equilibrium Baumol drag, and (39) makes precise what drives it. What pulls labour into local services is not the absolute AI shock to tradables but its

size *relative* to the AI shock to local services, scaled by μ : a frontier advance that raises tradable labour demand while leaving local productivity untouched raises the relative price of local services and, because demand for them is wage-elastic ($\chi_C > 0$), pulls labour in. If AI is productive inside local services too, the relative price moves less, or the other way. The factor μ on a_C is not a free parameter either: local productivity growth relaxes the labour constraint on the tradable block at the rate at which tradable employment responds to wages, so a small a_C offsets a much larger $\bar{\vartheta}$. Beyond $a_C = \bar{\vartheta}/\mu$ the Baumol channel does not merely weaken—it *reverses*, and labour flows out of local services on a frontier advance; on the non-empty interval $\bar{\vartheta}/\mu < a_C < \bar{\vartheta}L_T/\chi_C$, non-empty because $L_T/\chi_C - 1/\mu = (\mu L_T - \chi_C)/(\mu\chi_C) > 0$, the wage rises while local-service employment contracts. The unexposed-services benchmark is therefore not conservative for the paper’s central mechanism: it is the case in which that mechanism is strongest.

5.3 The AI development multiplier

For any pair of tradable-sector variables x_s and y_s , define the ξ -weighted covariance

$$\text{Cov}_\xi(x_s, y_s) \equiv \sum_{s \in S} \xi_s (x_s - \bar{x})(y_s - \bar{y}), \quad \bar{x} \equiv \sum_{s \in S} \xi_s x_s. \quad (41)$$

Theorem 1 (AI development multiplier). *Under Assumption 3, the effect of a marginal frontier AI improvement on capability growth is*

$$\partial_F \dot{H} = L_T \left[\text{Cov}_\xi(\psi_s(H, F), \vartheta_s(H, D)) - \frac{\chi_C}{\mu L_T - \chi_C} (\bar{\psi} - \psi_C(H, F))(\bar{\vartheta} - \mu a_C) \right] + \sum_{j \in S \cup \{C\}} L_j \partial_F \psi_j(H, F). \quad (42)$$

Equivalently, the composition part can be written

$$\partial_F \dot{H} = \sum_{s \in S} (\psi_s(H, F) - \psi_C(H, F)) L_s (\vartheta_s(H, D) - \mu \bar{\vartheta} + \mu(\mu\Lambda - 1)a_C) + \sum_{j \in S \cup \{C\}} L_j \partial_F \psi_j(H, F). \quad (43)$$

Under CES demand the Baumol coefficient is $\chi_C/(\mu L_T - \chi_C) = (1 - \sigma)L_C/(\mu - (1 - \sigma)L_C)$. The statement holds for any finite set S of tradable sectors, for arbitrary predetermined (H, D) , and for any demand system satisfying Assumption 3; under the unexposed-services benchmark $a_C = 0$ the Baumol term is proportional to $\bar{\vartheta}$ alone.

Proof. Differentiate (26) with respect to F at fixed (H, D) :

$$\partial_F \dot{H} = \sum_{s \in S} \psi_s \partial_F L_s + \psi_C \partial_F L_C + \sum_{j \in S \cup \{C\}} L_j \partial_F \psi_j.$$

The last sum is the learning-content term of (42), so it suffices to evaluate the composition part. labour-market clearing gives $\partial_F L_C = -\sum_s \partial_F L_s$, so the composition part equals

$$\sum_{s \in S} (\psi_s - \psi_C) \partial_F L_s,$$

and substituting (40), with $\mu\Lambda\Delta + \mu a_C = \mu\Lambda\bar{\vartheta} - \mu(\mu\Lambda - 1)a_C$, yields (43). Dividing and multiplying by L_T ,

$$\text{composition part} = L_T \sum_{s \in S} \xi_s(\psi_s - \psi_C)(\vartheta_s - \mu\Lambda\Delta - \mu a_C).$$

Expanding $\sum_s \xi_s(\psi_s - \psi_C)\vartheta_s = \text{Cov}_\xi(\psi_s, \vartheta_s) + (\bar{\psi} - \psi_C)\bar{\vartheta}$ and collecting the terms common to all sectors,

$$\text{composition part} = L_T \left[\text{Cov}_\xi(\psi_s, \vartheta_s) + (\bar{\psi} - \psi_C)(\bar{\vartheta} - \mu\Lambda\Delta - \mu a_C) \right] = L_T \left[\text{Cov}_\xi(\psi_s, \vartheta_s) + (\bar{\psi} - \psi_C)(1 - \mu\Lambda)\Delta \right],$$

using $\bar{\vartheta} - \mu a_C = \Delta$, and $1 - \mu\Lambda = -\chi_C\Lambda/L_T = -\chi_C/(\mu L_T - \chi_C)$ from (34) gives (42). No step used the cardinality of S , the relation between H and D , or any property of demand beyond (32) and Assumption 3. $\square$

First, a_C enters the Baumol term and *only* the Baumol term: an AI shock to local services is common across tradable sectors, so it cannot move a within-tradables covariance. The alignment statistic is therefore robust to how exposed local services are; the reallocation penalty is not. Second, the object that signs the Baumol penalty is the relative shock $\bar{\vartheta} - \mu a_C$, not $\bar{\vartheta}$. A country whose local service sector absorbs AI at more than $1/\mu$ of the average tradable rate has the Baumol term working *for* it. Since the maintained benchmark $a_C = 0$ is exactly the configuration that maximizes the penalty, any empirical implementation that sets it without evidence is choosing the sign of a first-order term by assumption. Section 9.3 states plainly that the public usage data cannot supply that evidence.

For future reference define the local AI development multiplier

$$\mathcal{M}(H, D, F) \equiv \partial_F \dot{H}(H, D, F). \quad (44)$$

Theorem 1 says that frontier AI affects development through exactly three channels. The *covariance term* asks whether AI-induced labour-demand gains are concentrated in high-learning tradable sectors. The *Baumol term* converts the *relative* labour-demand gain $\bar{\vartheta} - \mu a_C$ into a wage increase and thence into reallocation toward local services, at a rate governed by the single demand statistic χ_C . The *learning-content term* is the AI-specific channel: holding the allocation fixed, AI changes how much workers learn in each sector—negative where AI is used to take over the tasks through which juniors learned, positive where it tutors and codifies. The sign of $\mathcal{M}$ is therefore not the sign of the average labour-demand gain; it is the sign of an alignment, adjusted for the endogenous Baumol shift, plus a direct effect on the learning technology itself. Under the microfoundation (25), the learning-content term equals

$$\Psi'(F) g(D) \sum_{s \in S} L_s \psi_s a_s [(1 - d_s)\tau_s - d_s \eta_s] + L_C \partial_F \psi_C,$$

so a country can pass every composition test and still lose capability if it delegates learning-rich tasks outright (d_s above the mode threshold $\tau_s/(\tau_s + \eta_s)$)—or fail the composition test and still

gain, if AI-as-tutor raises ψ_s in the jobs that remain. This is the term the mode margin of Section 3.5 lives in: two countries with identical exposure a_s , identical readiness D , and identical sectoral composition can have learning-content terms of opposite sign purely because their d_s sit on opposite sides of that threshold. The factor $\Psi'(F)$ is common to the tradable sectors and therefore changes neither the sign of the term nor the relative sector contributions; it does mean the term's magnitude depends on where the frontier already is.

With two tradable sectors the covariance term takes the transparent form

$$\text{Cov}_\xi(\psi_s, \vartheta_s) = \xi_M \xi_P (\psi_P(H, F) - \psi_M(H, F)) (\vartheta_P(H, D) - \vartheta_M(H, D)). \quad (45)$$

The composition channel helps development only if the sector that is more capability-building also enjoys the larger AI-induced labour-demand gain. Note what (45) does *not* presume: it does not require $\psi_P \geq \psi_M$. The covariance is positive when the ladder ranking and the labour-demand ranking agree, whichever way round they run, and the empirically interesting configurations include the one in which frontier AI favors a tradable-services sector that turns out to teach *less* than manufacturing.

Corollary 1 (Why Bhagwati heterogeneity is essential). *Fix (H, D, F) with $S = \{M, P\}$ and shut down the learning-content channel, $\partial_F \psi_j \equiv 0$. Suppose $\psi_P = \psi_M > \psi_C$ and $\bar{\vartheta} > \mu a_C$. Then $\mathcal{M} < 0$.*

Proof. If $\psi_P = \psi_M$, (45) gives $\text{Cov}_\xi = 0$, and the learning-content term is zero by hypothesis. Since $\chi_C > 0$, $\bar{\psi} - \psi_C > 0$, and $\bar{\vartheta} - \mu a_C > 0$, the Baumol term in (42) is strictly negative. $\square$

Corollary 1 explains why the Bhagwati split is not cosmetic: if the tradable sectors are dynamically identical, an AI shock that favors tradables over local services only generates Baumol reallocation into local services. Note that the hypothesis is now $\bar{\vartheta} > \mu a_C$ rather than $\bar{\vartheta} > 0$: what generates the reallocation is the shock to tradables *relative* to local services. It also nests the two poles of the policy debate. *Premature deindustrialization* (Rodrik, 2016) is the case $\vartheta_M < 0$ at low readiness: frontier technology destroys manufacturing labour demand before capability has been built. *Blocked service-led escape*—the AI-era twist—is the case $\vartheta_P < 0$: the disembodied service exports celebrated by Bhagwati (1984) and Baldwin (2019) are exactly the tasks frontier AI automates, so the new rungs dissolve as they are grasped.

5.4 Trap regions

Definition 1 (AI development trap). *Given frontier level F , a state (H, D) is in the AI development trap if frontier AI raises average tradable labour demand at fixed wages relative to local services, $\bar{\vartheta} > \mu a_C$, but nevertheless lowers capability growth, $\mathcal{M} < 0$.*

The definition is intentionally strong. The trap is not the trivial case in which frontier AI reduces the country's tradable labour demand everywhere; it is the sharper case in which frontier AI is beneficial in the tradable block relative to local services, yet still pushes the economy away from

capability-building activity. Under the unexposed-services benchmark $a_C = 0$ the entry condition is simply $\bar{\vartheta} > 0$, which is how it is read throughout the discussion below; the general form is kept because it is the condition that actually delivers the Baumol reallocation, and because a country with AI-exposed local services can satisfy $\bar{\vartheta} > 0$ without being in the trap region at all.

Remark 8 (Drag versus trap). *Definition 1 is a local derivative condition, and a negative local derivative is not by itself a trap in the conventional multiple-equilibrium sense—no persistence, basin of attraction, or hysteresis is implied at this point. Until Section 7.2, the honest reading of $\mathcal{M} < 0$ is a development drag. We keep the name because Section 7.2 shows that the drag region is precisely what generates genuine traps—basins, discontinuous collapse, hysteresis—once the drift field folds, and because the welfare analysis of Section 6 shows the drag can be immiserizing on its own.*

Proposition 6 (Characterization of the trap). *Let $S = \{M, P\}$ and define*

$$\Gamma \equiv \xi_M \xi_P (\psi_P - \psi_M) (\vartheta_P - \vartheta_M) - \frac{\chi_C}{\mu L_T - \chi_C} (\bar{\psi} - \psi_C) (\bar{\vartheta} - \mu a_C) + \frac{1}{L_T} \sum_{j \in S \cup \{C\}} L_j \partial_F \psi_j(H, F), \quad (46)$$

with ψ_j evaluated at (H, F) and ϑ_s at (H, D) . Then $\mathcal{M} = L_T \Gamma$, and the state is in the trap at F if and only if $\bar{\vartheta} > \mu a_C$ and $\Gamma < 0$. If, under the diagonal convention (19) and for fixed F , $\Gamma(\cdot, F)$ is continuous and strictly increasing with $\Gamma(H_L, F) < 0 < \Gamma(H_H, F)$ for some $H_L < H_H$, then there is a unique threshold $H^(F) \in (H_L, H_H)$ with $\mathcal{M}(\cdot, F) < 0$ below and > 0 above.*

Proof. Immediate from Theorem 1, (45), $L_T > 0$, and the intermediate value theorem with strict monotonicity. $\square$

Remark 9 (State dependence and multiple trap regions). *If $\psi_P(H)$ is increasing and sufficiently convex—near zero for low H , as when learning from progressive services requires institutional infrastructure—or if the implementation share $g(D)$ is S-shaped along the diagonal, $\Gamma(\cdot, F)$ need not be monotone, and the economy can exhibit multiple trap regions and multiple local thresholds. The single-threshold case is a clean benchmark, not a general theorem. Section 7.2 shows that this multiplicity is not a pathology to be assumed away: it is what generates the model’s most consequential dynamics.*

6 Welfare: the trap is invisible in aggregate indicators

The trap of Definition 1 is a statement about capability, not welfare. This section shows, first, that in the trap every *aggregate* contemporaneous welfare indicator improves—wages, real income, consumption, and consumer-service employment all rise, so the trap cannot be seen in consumption data—and second, when the trap is nevertheless immiserizing in present-value terms. The claim is deliberately bounded: the trap is not literally invisible. Sectoral employment shares move on impact (Proposition 5), and export volumes and market shares move at a fixed wage (Proposition 2), which is what makes the empirical designs of Section 9 feasible. What the results below establish

is that the *conventional aggregate scoreboard* misclassifies trapped economies as AI winners; the information is in the composition, not the aggregates. This section works under the CES baseline, where homotheticity makes indirect utility equal to real income; Assumption 3 alone does not pin down a welfare metric.

Since u is homothetic, the indirect utility associated with u is real income,

$$V(H, F) = \frac{w}{P(q)}, \quad q = \frac{w}{A_C}, \quad (47)$$

where $P(\cdot)$ is the CES price index of (T, C) with $d \ln P / d \ln q = s_C$. Because primitive flow utility is $\ln u$, indirect flow utility is $\ln V$, and it is $\ln V$ —not V —that is integrated in the welfare functional below. Define the capability–wage feedback

$$\beta(H, F) \equiv \partial_H \ln \Omega(H, F) = (\mu - 1) \sum_{s \in S} \xi_s \partial_H \ln A_s(H, F) \geq 0. \quad (48)$$

Proposition 7 (Aggregate invisibility). *At any state,*

$$\partial_F \ln V = L_T \Lambda (\bar{\vartheta} - \mu a_C) + a_C, \quad \partial_H \ln V = L_T \Lambda \beta. \quad (49)$$

Hence if the state is in the AI development trap of Definition 1, a marginal frontier advance strictly raises the wage, strictly raises real income, and strictly expands consumer-service employment, while strictly lowering capability growth.

Proof. $\ln V = \ln w - \ln P(q)$ with $d \ln P / d \ln q = s_C$ and $\partial_F \ln q = \partial_F \ln w - a_C$, so $\partial_F \ln V = (1 - s_C) \partial_F \ln w + s_C a_C = L_T \partial_F \ln w + L_C a_C$ using $s_C = L_C$. Substituting $\partial_F \ln w = \Lambda \Delta + a_C$ from (38) and $L_T + L_C = 1$ gives the first identity in (49). For the second, A_C does not depend on H , so $\partial_H \ln q = \partial_H \ln w$ and $\partial_H \ln V = L_T \partial_H \ln w = L_T \Lambda \beta$ by (35) with $\partial_H \ln \Omega = \beta$. In the trap $\Delta = \bar{\vartheta} - \mu a_C > 0$, so $\partial_F \ln w = \Lambda \Delta + a_C > 0$, $\partial_F \ln V > 0$, and $\partial_F L_C = \chi_C \Lambda \Delta > 0$ by (39), while $\mathcal{M} < 0$ by definition. $\square$

The invisibility result therefore survives the introduction of AI into local services unchanged—but only because the trap’s entry condition was restated in terms of the relative shock Δ . What does not survive is the reading of that condition off tradable exposure alone. An economy with $\bar{\vartheta} > 0$ and $a_C > \bar{\vartheta} / \mu$ has rising wages and rising real income while consumer-service employment *contracts*; whatever else that economy is, it is not in the trap, and the compositional signature the trap predicts is absent. Mistaking the two is the practical cost of maintaining $a_C = 0$ without evidence.

Proposition 7 is the observational heart of the trap. In the trap region, frontier AI produces rising wages, rising real consumption, and a growing consumer-service sector—the standard scoreboard of successful development, and precisely the service-led growth pattern documented by Fan et al. (2023) and Peters et al. (2026). Consumption-based welfare comparisons will systematically classify trapped economies as AI winners; the divergence appears later, in capability-sensitive outcomes, while the contemporaneous evidence lives entirely in compositional data—which tradable sectors

shed employment, whose export sophistication stalls—that aggregate indicators average away. It is a statement about a representative agent: with two labour types (Appendix F), the skilled wage and aggregate income can rise while the unskilled wage falls, so the trap can leave a contemporaneous footprint in the skill premium and the earnings distribution even while the aggregates improve—and when the skill constraint binds, the same aggregate scoreboard can accompany an unambiguous capability loss (Corollary 3). And it is a statement about market income at world prices: where AI inputs are imported (Remark 6), the wage gains net of foreign rents are smaller than gross productivity suggests.

When is the trap actually bad, and not merely a benign exchange of future capability for present consumption? Consider an economy at a locally stable steady state H° of $\dot{H} = \mathcal{G}(H, F)$, where

$$\mathcal{G}(H, F) \equiv \sum_{s \in S} \psi_s(H, F) L_s(H, F) + \psi_C(H, F) L_C(H, F) - \delta H, \quad (50)$$

with local convergence speed $\kappa \equiv -\partial_H \mathcal{G}(H^\circ, F) > 0$, and evaluate welfare $W = \int_0^\infty e^{-\rho t} \ln V(H_t, F) dt$.

Proposition 8 (First-order immiserization condition). *Consider an unanticipated permanent marginal increase dF at a locally stable steady state H° . To first order,*

$$\frac{dW}{dF} = \frac{L_T \Lambda}{\rho} \left[\Delta + \frac{a_C}{L_T \Lambda} + \frac{\beta \mathcal{M}}{\rho + \kappa} \right] = \frac{L_T \Lambda}{\rho} \left[\Delta + \frac{a_C}{L_T \Lambda} + \frac{\kappa}{\rho + \kappa} \beta \frac{dH^\circ}{dF} \right], \quad (51)$$

with all objects evaluated at the steady state and $\Delta = \bar{\vartheta} - \mu a_C$. Hence, in the trap ($\Delta > 0 > \mathcal{M}$), the frontier advance lowers welfare if and only if

$$\beta |\mathcal{M}| > (\rho + \kappa) \left[\Delta + \frac{a_C}{L_T \Lambda} \right], \quad (52)$$

which under the unexposed-services benchmark $a_C = 0$ reduces to $\beta |\mathcal{M}| > (\rho + \kappa) \bar{\vartheta}$.

Proof. Linearizing $\dot{H} = \mathcal{G}(H, F + dF)$ around H° gives $dH_t = (\mathcal{M}/\kappa)(1 - e^{-\kappa t}) dF$, since $\partial_F \mathcal{G} = \mathcal{M}$ and $\partial_H \mathcal{G} = -\kappa$. To first order,

$$\frac{dW}{dF} = \int_0^\infty e^{-\rho t} \left[\partial_F \ln V + \partial_H \ln V \cdot \frac{dH_t}{dF} \right] dt = \frac{L_T \Lambda \Delta + a_C}{\rho} + L_T \Lambda \beta \frac{\mathcal{M}}{\kappa} \int_0^\infty e^{-\rho t} (1 - e^{-\kappa t}) dt,$$

using (49) with coefficients evaluated at the steady state. Since $\int_0^\infty e^{-\rho t} (1 - e^{-\kappa t}) dt = \kappa/[\rho(\rho + \kappa)]$, the first equality in (51) follows; the second uses $dH^\circ/dF = \mathcal{M}/\kappa$ (implicit function theorem on $\mathcal{G} = 0$). Condition (52) is the sign condition on the bracket. $\square$

The two forms of (51) read naturally. The first two terms are the permanent static gain—the relative labour-demand gain in tradables plus the direct consumer-price gain from cheaper local services. The second is the long-run capability loss $dH^\circ/dF = \mathcal{M}/\kappa$, converted into wages by the feedback β and discounted by the convergence-speed weight $\kappa/(\rho + \kappa)$. The trap is immiserizing—AI strictly lowers welfare while every aggregate contemporaneous *welfare* indicator improves—when

the economy is patient (low ρ), capability feeds strongly into tradable productivity (high β), and the compositional damage is large ($|\mathcal{M}|$ big) relative to the static gain ($\bar{\vartheta}$ small). Conversely, an impatient economy with weak capability–wage feedback rationally “sells the ladder”: the trap is then a growth trap but not a welfare trap. Equation (52) makes that dividing line sharp for the linearized experiment—it is a first-order condition around a stable steady state, not a global welfare theorem—and Section 8 shows that the $\beta\mathcal{M}/(\rho + \kappa)$ term is precisely the component of AI’s welfare effect that no private agent internalizes. Two qualifications carry over from Proposition 7: the criterion is representative-agent, so a passing grade does not establish that displaced tradable workers gain; and it is national, so a terms-of-trade gain for the home country is partly a transfer from its trading partners rather than a global gain.

7 Dynamics with a moving frontier

This section is where the model needs a one-dimensional state, and so where the diagonal convention (19) is imposed: readiness is tied to capability by $D = d(H)$ with d continuous and nondecreasing, nesting $D \equiv H$, and $\mathcal{G}(H, F) \equiv \mathcal{G}(H, d(H), F)$. Everything before this section holds for arbitrary predetermined (H, D) . Appendix E gives the planar system that the convention stands in for, together with the two cases—readiness indexed to development, or readiness quasi-fixed over a capability transition—under which the reduction is legitimate, and the case it excludes: readiness moving fast enough to shift the drift field within a transition. The dynamic system is

$$\dot{H}_t = \mathcal{G}(H_t, F_t), \quad \dot{F}_t = \varepsilon > 0, \quad (53)$$

where ε is the speed of the world AI frontier and $\mathcal{G}$ is the drift field (50), assumed continuously differentiable on $\mathcal{K} \times \mathbb{R}_+$.

7.1 Tracking a regular branch

Assumption 4 (Regular branch). *For every $F \geq 0$ there is a unique $H^\circ(F) \in \mathcal{K}$ with $\mathcal{G}(H^\circ(F), F) = 0$; there is $\kappa > 0$ with $\partial_H \mathcal{G}(H, F) \leq -\kappa$ on the relevant state space; and $B \equiv \sup_{F \geq 0} |\mathrm{d}H^\circ / \mathrm{d}F| < \infty$.*

Proposition 9 (Moving-frontier transition dynamics). *Under Assumption 4, the instantaneous steady-state locus satisfies*

$$\frac{\mathrm{d}H^\circ(F)}{\mathrm{d}F} = -\frac{\mathcal{M}(H^\circ(F), F)}{\partial_H \mathcal{G}(H^\circ(F), F)}, \quad (54)$$

so the locus slopes upward in F exactly where the development multiplier is positive and downward exactly where it is negative. Moreover, the tracking error $E_t \equiv H_t - H^\circ(F_t)$ satisfies

$$|E_t| \leq e^{-\kappa t} |E_0| + \frac{\varepsilon B}{\kappa} (1 - e^{-\kappa t}). \quad (55)$$

Proof. The implicit function theorem on $\mathcal{G}(H^\circ(F), F) = 0$ gives $\mathrm{d}H^\circ / \mathrm{d}F = -\partial_F \mathcal{G} / \partial_H \mathcal{G}$, and

$\partial_F \mathcal{G} = \mathcal{M}$ by (44), yielding (54). For the bound, $\dot{E}_t = \mathcal{G}(H_t, F_t) - \mathcal{G}(H^\circ(F_t), F_t) - \varepsilon \, dH^\circ / dF$; the mean value theorem gives $\mathcal{G}(H_t, F_t) - \mathcal{G}(H^\circ(F_t), F_t) = \partial_H \mathcal{G}(\tilde{H}_t, F_t) E_t$ for some intermediate $\tilde{H}_t$, so $\frac{d}{dt} |E_t| \leq -\kappa |E_t| + \varepsilon B$, and Gronwall's inequality yields (55). $\square$

Proposition 9 gives the local multiplier a transition-dynamics interpretation: it is the slope of the moving steady-state locus, up to the stable denominator. For a slowly moving frontier, the economy stays within an $O(\varepsilon)$ band of the locus, so phase-diagram language is legitimate.

7.2 Folds: collapse, escape, and hysteresis

Assumption 4 imposes a globally unique, uniformly stable locus, which rules out by fiat exactly the configurations that Remark 9 identifies as natural: S-shaped absorption or convex ladder weights generate multiple stable branches. This subsection analyzes that case. The economics are stark: when branches fold, *continuous* frontier progress produces *discontinuous* development outcomes, and the multiplier's sign along each branch—by (54), the direction of the branch's drift—determines who tips which way.

Remark 10 (Where the multiplicity comes from). *The S-shape of the implementation share $g(D)$ is not a free curvature assumption; fixed adoption costs deliver it under mild conditions. Suppose implementing an arriving AI opportunity requires a firm-level fixed cost c drawn from a distribution with cdf Π and unimodal density π (the symbol κ is reserved for the convergence speed), and yields a private gain $b(H)$ increasing in capability (complementary skills, data, organizational practices). Firms implement iff $b(D) \geq c$, so the implementation share is $g(D) = \Pi(b(D))$ and*

$$g''(H) = \pi'(b(H)) [b'(H)]^2 + \pi(b(H)) b''(H).$$

When b is affine the second term vanishes and g inherits the shape of π exactly: convex where the cost density rises, concave where it falls—an S-shape from primitives. For nonlinear b a unimodal cost density is not by itself enough; the S-shape additionally requires that the curvature of b not overturn the sign of the first term, which is a joint restriction on π and b rather than a property of the cost distribution alone. Data and network externalities in adoption, minimum efficient scale in tradable-service clusters, or coordination between skills and AI investment generate the same shape with genuinely strategic multiplicity layered on top; convexity of $\psi_P(H)$ has the analogous microfoundation in institutional prerequisites for learning from progressive services.

First, the object that folds is the capability drift field $\mathcal{G}(H, F)$ of (50), not the adoption function: g reaches $\mathcal{G}$ only indirectly, through α_s and hence ϑ_s and Ω_s , and—under (25)—through the learning weights. Second, an S-shaped g therefore only can induce a folded drift field: it does so when the capability feedback is strong enough that $\mathcal{G}(\cdot, F)$ crosses zero three times, and not otherwise. Curvature in adoption is a source of the required nonmonotonicity, not a proof of it, so Assumption 5 is a restriction imposed on the drift field rather than derived from primitives. The quantitative question—whether the folds sit in economically relevant regions of (H, F) —is exactly the kind of restriction the estimated model of Section 9 is designed to check.

Assumption 5 (Fold structure). *There exist $0 < \hat{F} < \infty$ and continuous functions*

$$H_{\text{lo}} : \mathbb{R}_+ \rightarrow \mathcal{K}, \quad H_{\text{mid}}, H_{\text{hi}} : [0, \hat{F}) \rightarrow \mathcal{K},$$

with $H_{\text{lo}}(F) < H_{\text{mid}}(F) < H_{\text{hi}}(F)$ on $[0, \hat{F})$, such that:

- (i) *for $F < \hat{F}$, the zero set of $\mathcal{G}(\cdot, F)$ on $\mathcal{K}$ is $\{H_{\text{lo}}(F), H_{\text{mid}}(F), H_{\text{hi}}(F)\}$, with $H_{\text{hi}}(F) - H_{\text{mid}}(F) \rightarrow 0$ as $F \uparrow \hat{F}$; for $F > \hat{F}$, the unique zero is $H_{\text{lo}}(F)$;*
- (ii) *(uniform stability away from the fold) for every $\eta > 0$ there exist $r_\eta, \kappa_\eta > 0$ such that $\partial_H \mathcal{G}(H, F) \leq -\kappa_\eta$ whenever $|H - H_{\text{hi}}(F)| \leq r_\eta$ and $F \leq \hat{F} - \eta$, or $|H - H_{\text{lo}}(F)| \leq r_\eta$ and $F \geq 0$;*
- (iii) *(uniform decay beyond the fold) for every $c > 0$ there is $\gamma_c > 0$ such that $\mathcal{G}(H, F) \leq -\gamma_c$ whenever $F \geq \hat{F} + c$ and $H \geq H_{\text{lo}}(F) + c$;*
- (iv) *H_{lo} is differentiable with $\sup_{F \geq 0} |dH_{\text{lo}}/dF| \leq B < \infty$ and, for each $\eta > 0$, H_{hi} is differentiable on $[0, \hat{F} - \eta]$ with $\sup_{F \leq \hat{F} - \eta} |dH_{\text{hi}}/dF| \leq B_\eta < \infty$.*

Proposition 10 (Collapse of the upper branch). *Let Assumption 5 hold and start the economy on the upper branch: $H_0 = H_{\text{hi}}(F_0)$ with $F_0 < \hat{F}$. For every sufficiently small $c > 0$ there exists $\bar{\varepsilon} > 0$ such that for all $\varepsilon \in (0, \bar{\varepsilon})$:*

- (i) *$|H_t - H_{\text{hi}}(F_t)| \leq c$ for all t with $F_t \leq \hat{F} - c$;*
- (ii) *there is $T_c < \infty$, independent of ε , such that $|H_t - H_{\text{lo}}(F_t)| \leq c$ for all $t \geq t_c + T_c$, where $t_c \equiv (\hat{F} + c - F_0)/\varepsilon$.*

In particular, as $\varepsilon \downarrow 0$ the economy tracks the upper branch until the frontier reaches $\hat{F}$ and then transits to the lower branch during a vanishing interval of frontier progress: capability collapses discontinuously in F .

Proof. Appendix A. □

Corollary 2 (Escape waves). *Suppose instead the lower and middle branches merge at a fold $\check{F}$ (with the symmetric counterparts of conditions (i)–(iv)), which by (54) is the natural configuration when $\mathcal{M} > 0$ along the lower branch. Then an economy starting on the lower branch tracks it until the frontier reaches $\check{F}$ and then transits to the upper branch: frontier progress dissolves the trap and produces a discontinuous take-off.*

Which scenario obtains is governed by the paper’s sufficient statistic: by (54), each branch drifts in the direction of $\mathcal{M}$ along it. If the covariance-minus-Baumol object is negative along the upper branch, frontier progress pushes high-capability economies toward the fold—the escalator is dismantled under their feet (Figure 3). If it is positive along the lower branch, frontier progress erodes the trap from below. The frontier does not permit indifference: as F grows without bound, every fold within reach is eventually crossed.

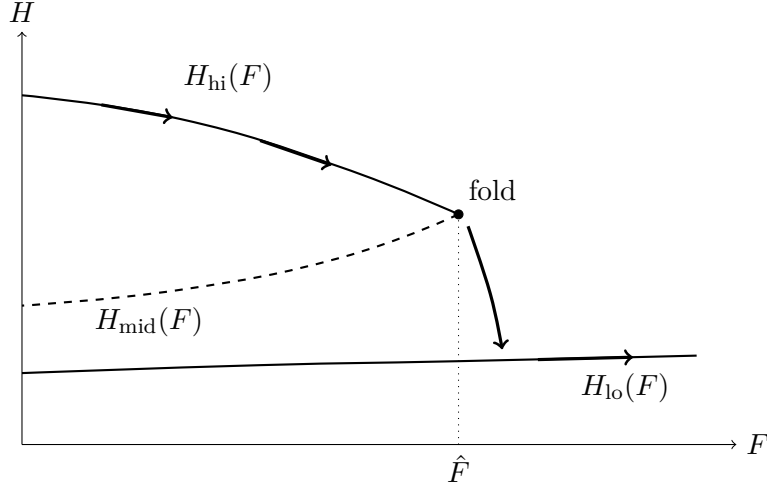


Figure 3: Collapse of the upper branch (Proposition 10). Solid curves are stable branches of $\mathcal{G}(\cdot, F) = 0$; the dashed curve is the unstable middle branch. The upper branch slopes downward because $\mathcal{M} < 0$ along it (equation (54)). A slowly moving frontier carries the economy along the upper branch to the fold at $\hat{F}$, after which capability transits rapidly to the lower branch.

Remark 11 (Hysteresis and the big push). *The collapse in Proposition 10 is irreversible in F : halting—or partially reversing—frontier progress does not restore high capability. The reason depends on where the frontier ends up, and the distinction has policy content. Beyond the fold ($F > \hat{F}$), Assumption 5(i) leaves $H_{lo}(F)$ as the only steady state: there is no middle branch to cross and no upper branch to reach, so a one-off increase in H alone cannot restore a high steady state under the stated drift field—the economy simply drifts back down to H_{lo} . Recovery requires either (a) the frontier is reversed far enough below $\hat{F}$ that the middle and upper branches re-emerge, after which a discrete push lifting H above the re-emergent H_{mid} restores the upper branch—this is the big push of Murphy et al. (1989) in its literal form; or (b) a non-marginal policy intervention shifts the drift field $\mathcal{G}$ itself enough to recreate a high branch at the prevailing frontier, followed, if the intervention leaves the economy below it, by a push across the new unstable threshold. What survives in both cases is the asymmetry that motivates early action: marginal, temporary ladder subsidies that would have sufficed to hold the economy on the upper branch before $\hat{F}$ —by shifting $\mathcal{G}$ slightly and postponing the fold—accomplish nothing after it, because past the fold a marginal shift no longer changes the number of steady states. The policy window is open only while the branch exists, and Proposition 10 says the frontier closes it in finite time. The closing date itself is not knowable *ex ante*—it depends on the speed of the frontier and the location of the fold, both uncertain—but its existence converts uncertainty about timing into an argument for acting early rather than an argument for waiting.*

8 Efficiency and policy

Capability accumulation (26) is external to all private decisions, so the competitive equilibrium is generically inefficient. In addition, the country as a whole faces downward-sloping residual demand

(10) while individual firms price-take, so there is a standard terms-of-trade externality. This section characterizes the constrained planner and shows that optimal policy combines the two corresponding instruments—an export tax and ladder subsidies—which pull in opposite directions on the tradable block.

The planner chooses the labour allocation $\{L_j(t)\}_{j \in S \cup \{C\}}$, sells tradables along the residual demand curves, and spends the proceeds on imports. Throughout this section we write $\psi_j(H)$, suppressing the F argument: the frontier is exogenous to the planner, and the learning-content channel changes the level of the shadow value λ without altering the structure of the wedges. The problem is:

$$\max \int_0^\infty e^{-\rho t} \ln u(T_t, C_t) dt \quad \text{s.t.} \quad T = \sum_{s \in S} \Xi_s(F) (A_s(H, F) L_s)^{1-\frac{1}{\mu}}, \quad C = A_C L_C, \quad (56)$$

labour-market clearing (28), and the capability law (26).

Proposition 11 (Planner wedges and decentralization). *Along an interior optimum there exist a co-state $\lambda_t \geq 0$ (with $\lambda_t > 0$ whenever $\partial_H \ln A_s > 0$ for some s with positive employment at some future date) and a multiplier ϖ_t such that*

$$\frac{u_T}{u} \left(1 - \frac{1}{\mu}\right) p_s A_s + \lambda \psi_s(H) = \varpi \quad (s \in S), \quad \frac{u_C}{u} A_C + \lambda \psi_C(H) = \varpi, \quad (57)$$

where $p_s = \Xi_s Y_s^{-1/\mu}$ at planner quantities. The optimum is decentralized by

- (i) a uniform ad valorem export tax at rate $1/\mu$, and
- (ii) employment subsidies $\tau_j = \tilde{\lambda} \psi_j(H)$ per worker in sector j , where $\tilde{\lambda} \equiv \lambda u/u_T$ is the shadow value of capability in units of the tradable numeraire,

both financed lump-sum. Only the differentials $\tau_j - \tau_C = \tilde{\lambda}(\psi_j - \psi_C)$ matter for allocations.

Proof. Appendix A. □

First, the headline instrument is the ladder subsidy, and the model does not rationalize export promotion. The dynamic wedge subsidizes employment in high- ψ sectors: pay for learning, wherever learning occurs. The static wedge is an export tax—the country has market power $1/\mu$ on its residual demand curves and atomistic exporters over-export relative to the optimum—but this is a theoretical benchmark rather than practical advice: as $\mu \rightarrow \infty$ (a truly atomistic country, the empirically relevant case for most developing economies), the terms-of-trade wedge vanishes and optimal policy reduces to the pure ladder subsidy $\tilde{\lambda}(\psi_j - \psi_C)$. For interior μ the net stance toward tradables is ambiguous, depending on $\tilde{\lambda} \psi_s$ against $p_s A_s/\mu$; what is unambiguous is the cross-sector structure of the subsidy.

Second, the development multiplier is the policy-relevant externality. In the welfare decomposition (51), the term $\beta \mathcal{M}/(\rho + \kappa)$ is exactly the component of AI's welfare effect operating through the

uninternalized state variable: private agents reallocate in response to AI as if $\psi_j \equiv 0$, and the planner’s valuation of that reallocation is $\tilde{\lambda} \partial_F \dot{H} = \tilde{\lambda} \mathcal{M}$ at the margin. Measuring $\mathcal{M}$ is therefore not only a positive exercise; it prices the wedge.

Third, timing interacts with the dynamics of Section 7.2. While the economy is on the upper branch, marginal ladder subsidies shift the branch and can postpone or prevent the fold. After a collapse at a frontier level beyond the fold, no marginal subsidy restores high capability, and—because the upper and middle branches no longer exist—neither does a discrete push in H on its own: restoration requires either a retreat of the frontier below $\hat{F}$ or a non-marginal shift of the drift field, in each case followed by a push past the recreated unstable branch (Remark 11). The same instrument is cheap ex ante and enormously expensive—possibly unavailable—ex post: a precise sense in which AI raises the option value of acting early.

Fourth, the mapping from the abstract subsidy to implementable instruments runs through the model’s own decomposition. Because the externality attaches to *learning*, not to sectors or exports per se, the implementable counterparts are instruments that raise $\psi_j L_j$ or $g(D)$ directly: AI-adoption and worker-training subsidies, digital and compute infrastructure, buyer-linkage and supplier-development programs, and support for task upgrading within firms rather than sectoral employment targets; where AI inputs are imported (Remark 6), the taxation and negotiation of foreign AI rents is a separate instrument aimed at the division of surplus rather than the allocation of labour. The first caveat is informational: the optimal subsidies depend on $\psi_j(H, F)$, which is estimated with error and is itself potentially endogenous to policy, so the subsidies should be read through a robust-policy lens—their cross-sector *ranking* is more identified than their level. The second is jurisdictional: the planner here takes the frontier as given. If the direction of frontier innovation is chosen abroad and imposes capability or terms-of-trade losses on developing economies, national ladder subsidies mitigate but cannot correct the international externality—the distinction Korinek and Stiglitz (2021) draw between domestic redistribution and internationally generated losses—and the relevant instruments become international: participation in standard-setting, compute and model access agreements, and the direction-of-innovation itself.

9 Taking the development multiplier to data

Theorem 1 reduces the development effect of AI to measurable objects: employment shares (ξ_s, L_C) , the demand statistic χ_C , elasticities (μ_s, ζ_s) , ladder weights $\psi_s(H, F)$, the local-service exposure a_C , and fixed-wage labour-demand effects $\vartheta_s = (\mu - 1)\alpha_s(H, D) - \mu\nu_s$. Write the empirical counterpart of the multiplier’s *composition* part for country i at time t as

$$\widehat{\mathcal{M}}_{it}^c = \sum_{s \in S} (\widehat{\psi}_s(H_{it}) - \widehat{\psi}_C(H_{it})) L_{ist} (\widehat{\vartheta}_{ist} - \mu \widehat{\Lambda}_{it} \widehat{\vartheta}_{it}), \quad (58)$$

where L_{ist} is sector s ’s *share of country i ’s employment* in period t (in the model labour is normalised to one, so sectoral labour and its share coincide; in a country panel they do not, and the share is

the object that belongs in (58)).

First, the pass-through term $\mu \hat{\Lambda}_{it} \bar{\vartheta}_{it}$ is the *common-residual-elasticity benchmark*: it presumes a single μ across tradable sectors. When residual demand elasticities differ, Proposition 12 shows the wage response becomes $L_T \bar{\vartheta} / (L_T \bar{\mu} - \chi_C)$ and the multiplier acquires an elasticity-alignment term in $\text{Cov}_\xi(\psi_s, \mu_s)$ —whether capability-building sectors also face more elastic world demand. Equation (58) is therefore the specification under the common- μ benchmark, and the heterogeneous- μ_s version of Appendix C is the preferred quantitative specification, with μ_M and μ_P estimated separately. Second, the superscript c is doing work: (58) omits the learning-content term $\sum_j L_j \partial_F \psi_j$ of Theorem 1, which requires task-level data on what AI-assisted workers stop and start learning—a firm- or worker-level object rather than a country–sector one—and which can reverse the sign of the full multiplier. Everything below that uses $\hat{\mathcal{M}}^c$ tests the composition channel, and tests the full development effect only under the maintained condition that the learning-content term is zero, separately controlled, or orthogonal to $\hat{\mathcal{M}}^c$.

Estimating every object in (58) from the same post-2022 sample would invite generated-regressor and endogeneity problems: capability proxies would be fitted to the very reallocation they are supposed to predict. The design below therefore separates the estimation of learning effects from the estimation of AI shocks, and treats the model’s four predictions as tests of an out-of-sample construction rather than in-sample fit. A full execution is the task of a companion empirical paper; this section fixes the architecture that the theory implies, and Section 9.3 states what the public frontier-AI usage data behind Section 2 can and cannot discipline within it.

9.1 Ladder weights from pre-AI variation

The ladder weights ψ_s should be estimated on historical data, before frontier AI is in the picture, and then carried out of sample. The regression object is the causal effect of sectoral activity on subsequent capability, in the spirit of Atkin et al. (2026).

The preferred design is a destination-driven foreign demand shift-share. Country i ’s sector- s shock is built from pre-period destination shares interacted with destination-specific expenditure growth in that sector, excluding the home country’s own contribution to each destination and using leave-one-out destination growth net of any home-country supply effect; identification then rests on one exclusion restriction, stated explicitly: conditional on country–time effects, sector–time effects, and pre-period sectoral composition, destination expenditure growth affects a country’s subsequent capability outcomes *only* through the sectoral labour it induces. The restriction is testable in the usual ways—pre-trends in the outcome, balance of the shock against initial capability and readiness, and placebo shocks constructed from destinations that do not buy the sector—and it is violated in specific, checkable directions (destination-specific product standards travelling with demand; destination composition correlated with subsequent immigration or investment ties).

The alternatives are not interchangeable substitutes for that design, and the paper does not treat them as a menu. Multinational buyer entry plausibly transmits technical knowledge directly, which is a direct violation when the endogenous variable is sectoral employment alone; digital

connectivity raises capability and information access directly (Hjort and Poulsen, 2019); trade-agreement accession moves imported inputs, competition, investment, and buyer relationships simultaneously; and transport infrastructure alters firm survival and market access independently of employment. Each can serve as a robustness instrument only under an explicitly stated additional restriction, or as an over-identification check whose disagreement with the demand shift-share is informative about which channel is operating.

First, the dependent variable must not be mechanically constructed from the same sectoral shares that appear on the right-hand side—export-complexity indices risk exactly this circularity—so the capability outcomes should be entry into more complex products and services, export quality, occupational upgrading, diversification across products and destinations, and the survival and growth of exporting firms. Second, the estimation should test the learning *basis*, but carefully: the baseline law (26) predicts that employment composition causes future capability conditional on output, productivity, and export scale, whereas the output-based alternative of Appendix D predicts the conditional effect vanishes—and conditioning on output, productivity, and export scale means conditioning on mediators that the same employment shock produces, which biases the comparison in an unknown direction. The cleaner version instruments employment and output *separately*—for instance with variation in labour intensity, or in labour-market institutions interacted with the demand shock, that moves worker-years while holding production scale relatively fixed—and reports the two first stages. The experimental evidence in Atkin et al. (2017), where exporting improved quality and technical efficiency through knowledge transferred from foreign buyers, suggests that export relationships are a live candidate basis, not a straw man. Third, $\hat{\psi}_s$ enters (58) as a generated regressor, so split-sample or cross-fitted construction handles overfitting but not first-stage sampling uncertainty: inference on $\widehat{\mathcal{M}}^c$ requires re-estimating $\hat{\psi}_s$, $\hat{\vartheta}_s$, the elasticities, and hence $\widehat{\mathcal{M}}^c$ itself inside each bootstrap or cross-fitting replication, rather than treating the plug-in construction as data.

9.2 Separating domestic absorption from foreign displacement

The effect ϑ_{is} nets a domestic absorption component against a foreign competition component, and a generic occupational-exposure index interacted with the ChatGPT release identifies neither. The domestic component calls for predetermined sectoral AI exposure interacted with predetermined country readiness—digital infrastructure, electricity reliability, cloud and API access, language coverage, worker skills, pre-existing software intensity—with sharper designs exploiting staggered model or language availability, submarine-cable and data-center openings, changes in cross-border licensing, or firm-level rollout dates. The foreign component can be built as a shift-share with pre-AI competitor weights: home country i ’s exposure to foreign AI competition in sector s is the pre-period share of its competitors in each destination interacted with those competitors’ AI shocks, with destination–sector–time fixed effects separating foreign supply competition from common demand shifts. Proposition 2 then does real empirical work: the responses of export prices, quantities, revenues, market shares, and employment are *different objects with different signs*, and

reporting them separately is what distinguishes a country losing competitiveness (market share falling: absorption too low) from a country riding inelastic world demand down (market share stable, employment falling: $\zeta_s \leq 1$). Two identification caveats from the theory: α_s and ν_s are not separately identified from sectoral employment responses alone, which is why the price and market-share margins are essential; and the scale of F is arbitrary, so implementation requires a normalization making α_s , ν_s , and $\mathcal{M}$ comparable across sectors and countries.

9.3 What the public usage data can and cannot discipline

Most of the architecture above waits on employment, trade, demand, and learning data that must be assembled. One block can be implemented now: large-scale public evidence on who uses frontier AI, for which tasks, and under which division of labour already exists, and Section 2 reported the three facts from it that motivate the model. This subsection reports the release-data implementation and does the accounting those facts require: which model object each observable is an analogue of, which it is not, what the descriptive regressions behind Fact 3 survive, and which data request remains binding.

Table 1 states the mapping discipline. Its middle column is deliberately weaker than a parameter name: each AEI observable has a closest empirical analogue, and the third column records what that analogue does not identify. The distinction is not pedantic. Observed usage is an equilibrium object, so a usage share can move with model capability, product design, the mix of users and tasks, migration between the chat products and the API, and classifier revisions—all of which the release-specific harmonization matrix documents—and none of which is the structural object the model defines.

Table 1: Anthropic Economic Index observables, their closest empirical analogues, and what they do not identify.

AEI observable	Closest empirical analogue	What it does not identify
Anthropic Usage Index (AUI)	relative intensity of observed Claude conversations per working-age person; a noisy ordering of realised use and of the readiness stock D that governs $g(D)$	the implementation share g : the index is not bounded above, does not separate users from conversations per user, is not normalised by a country’s task opportunity set, and is relative to the global distribution of Claude use

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AEI observable	Closest empirical analogue	What it does not identify
Usage-income elasticity	a positive, non-flattening income gradient in observed use over the range covered	the shape or saturation point of g ; the two elasticities come from different samples, income ranges, and institutions, so their gap alone does not measure the weight of non-income determinants
Concentration	divergence between scale and per-capita intensity early in diffusion	persistence: two methodologically non-identical snapshots, published under changing products, users, and classifiers
Directive share of chat conversations	delegation-mode composition of observed interactions at particular dates and on particular products	the delegation profile $\Psi(F)$: a mode share is bounded mechanically, which is not evidence for the economic boundedness Assumption 2 imposes; and calendar time is not frontier units
Automation tilt	the conditional mode share d_s covarying with observed usage intensity <i>within broad task families</i> ; the change specifications ask whether the association survives country heterogeneity	why, or a response to F : user mix, product mix, selective cell publication, within-GWA task heterogeneity, and complementary capability are not separated
Work orientation, May 2026 (computed): coursework share -5.17 pp per log point of AUI (0.55) raw, $+0.15$ (0.62) adjusting for job-category composition; work-use share $+1.29$ (0.52) raw, $+0.32$ (0.49) adjusted	raw composition differences in what low- and high-usage countries bring to the model	depth of absorption: the gradient is not separable from published task composition, and the direct work-use outcome carries no robust gradient
Education level of prompts rises $\approx$ one school-year per year of use, tapering after a year (March 2026 report)	user learning, selection into sustained use, or task upgrading among continuing users	the learning parameters τ_s, η_s or the mode share d_s : the learning content of the tasks AI takes over at work, relative to the sector mean, is a within-worker object

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AEI observable	Closest empirical analogue	What it does not identify
1P API top-ten task share	task composition of Anthropic first-party API traffic, its concentration within a vintage, and sensitivity to the unit of weighting	what world demand delegates to the frontier: one vendor’s own API, excluding third-party hosting and other providers, with no buyer geography; the June break prevents a common concentration trend
No split of observed usage between tradable and non-tradable activity; the export-relevant task triple is 37.5% of world chat usage, and the remainder is unassigned	the breadth of frontier use across the whole task space	the relative exposure of tradable and local consumer services, and hence the local-service exposure a_C on which the sign of the Baumol term turns (Theorem 1); the benchmark $a_C = 0$ is a maintained restriction, not a finding
Coverage is language-model interaction: chat, Cowork, and 1P API text traffic	the diffusion of generative AI as an interactive tool	the manufacturing channel: AI embodied in robotics and process control does not appear, so these data are close to silent on a_M and cannot motivate the M block
<i>Sources:</i> Handa et al. (2025) (first release); Anthropic (2025) (September 2025 report and release); Anthropic (2026a) (March 2026 report); Anthropic (2026b) (June 2026 report: calendar-month aggregates and feed separation); Anthropic (2026c) (data releases). Release-data calculations use the pinned public snapshot described in the text; connector-era May values are retained where labelled. The AUI is a country’s share of observed Claude usage divided by its share of the world’s working-age population; 1.0 means usage proportional to population. The granular primary sample is 76 countries with balanced GWA publication in both April and May, selected from country-overall cohorts of 114 and 121. Missing cells are unpublished, never zero. Every result is a descriptive relationship between equilibrium usage outcomes, with no causal content.		

Absorption and readiness. The AUI provides a noisy ordering of relative usage intensity across countries. Conditional on task opportunities and access, it is suggestive of the readiness gradient underlying $g(D)$, where implementation sits in readiness and learning in capability; it does not measure the implementation share itself, and its level cannot be read as a position relative to g ’s upper limit. It is also a *relative* index, which rules out a further use: because the denominator moves with global usage, a country’s absolute use can rise while its index is flat or falls, so the series cannot serve as a cardinal time-series counterpart of g , still less of the frontier state F . What the data do support is an ordering—steeply increasing in income, with the poorest countries near the bottom of the observed range—and a warning about levels, since the top of the ranking churns across snapshots. The income gradient of the September 2025 report is consistent with a strictly increasing g over the observed range, and the fact that the within-US-state elasticity (1.8) exceeds

the cross-country one (0.7) is often read as evidence that non-income factors carry the cross-country gradient. That inference does not follow from the two coefficients alone: the samples differ in income range, institutions, and the distribution of the dependent variable. The relevant evidence is the residual variation within each regression, which the public aggregates do not report—and which the readiness stock D of Appendix E is the model’s device for absorbing.

Delegation-mode composition. The share of chat conversations classified as directive rose from 27 to 39 percent over the eight months separating the first and third releases (Anthropic, 2025), and the coarser automation/augmentation binary is 48.6/51.4 in the May 2026 aggregate. It is tempting to read those points as tracing the delegation profile $\Psi(F)$ that Section 3.5 requires to be bounded and increasing. The release audit shows why the correspondence is too quick: legacy vintages and June differ in classifier, sampling, calendar period, or product universe, and only April–May 2026 qualifies for a common panel. In the model, Ψ governs the cumulative intensity of learning-relevant tasks delegated as the frontier advances; a mode share is the fraction of observed interactions assigned to a classifier bucket, and it moves with product design, user and task mix, migration between chat, Claude Code, and the API, and call architecture. That a share is mechanically bounded in $[0, 1]$ is not evidence for the economically substantive boundedness imposed on Ψ . Even a harmonized calendar series would require a separate frontier-capability measure to convert time into frontier units. The observed mode series is therefore a description of interaction composition, not an estimate of the profile.

Learning content. Nothing in the public aggregates identifies the learning parameters τ_s and η_s of (25), which are within-worker objects. The mode share d_s is closer to something the data describe, but only in a limited sense: the published automation share is a share of observed *conversations*, whereas d_s is a share of implemented *production tasks* in a sector, so the cross-country gradient of Fact 3 is suggestive of how d_s varies with readiness and silent on its level. One fact bears on its plausibility: the March 2026 report finds that the education level required to understand user prompts rises by almost a school-year per additional year of Claude use, tapering after roughly a year, and that experienced users achieve higher task-success rates (Anthropic, 2026a). Even taken as causal, that would not sign η_s , which is the learning intensity of the tasks AI takes over relative to the sectoral mean. Rising prompt complexity among longer-tenured users is consistent with users learning the tool, with selection into sustained use, and with task upgrading unaccompanied by any change in workers’ productive capability; the report itself stresses that its sample mixes cohorts of very different vintages. The tenure gradient establishes the AI-as-tutor mechanism as empirically plausible—a live $\tau_s > 0$ operating through complementary use, consistent with the experience gradient in Brynjolfsson et al. (2025)—without identifying the sign of the net learning-content rate $(1 - d_s)\tau_s - d_s\eta_s$. The model leaves that net rate free for exactly this reason, and the firm- and worker-level design of Section 9.5 remains the identifying design.

The estimand ladder. The exact target remains the within-leaf-task regression

$$\text{Automation}_{ikt} = \beta \ln \text{AUI}_{it} + \lambda_k + \delta_t + X_i' \gamma + \varepsilon_{ikt}, \quad (59)$$

with k a leaf task and the country’s mode composition observed within it. No public vintage carries country mode within leaf task: August, November, and February carry leaf-task shares but only country-overall mode, while April–May carry mode at GWA, request-major, and SOC-major resolution. The finest feasible direct specification therefore replaces k by one of 35 GWAs. The primary estimate pools April and May, absorbs GWA and month fixed effects, controls only for log 2024 GDP per capita and the published work-use share, gives each country equal total weight, and clusters two ways on country and GWA. It uses the 76 countries with balanced country–GWA support—a median of 20 published GWAs per country, not all 35—and 3,114 cells; the estimating rule excludes unclassified observations from the mode denominator, though on this sample every published cell is fully classified so the exclusion does not bind. The coefficient is -0.99 percentage points per log AUI (se 0.40, 95 percent CI $[-1.77, -0.22]$); a 999-draw Webb wild bootstrap on the 35 GWA clusters gives $p = 0.017$. Equal-cell weighting gives -0.98 $[-1.75, -0.21]$, while traffic weighting—the average observed conversation rather than the average country—gives -2.13 $[-4.19, -0.07]$.

The change specifications are weaker and more revealing. Adding country fixed effects gives $+0.41$ $[-3.28, 4.09]$, and first-differencing April from May gives $+0.71$ $[-3.12, 4.54]$ over 1,557 country–GWA changes; both contain zero and the reversal region and neither is informative enough to recover a frontier derivative—there are only two months, no untreated group, AUI itself is relative, and classifier or common product shocks remain. A country-aggregate change specification over the same two months, using the full 114-country cohort rather than the balanced GWA panel, is more precise and gives -3.02 $[-5.43, -0.61]$: negative and significant, unlike the two within-GWA change estimates, though it is a country-level average rather than a within-task result and rests on only two adjacent months.

None of the four outcomes fixed before estimation—robust gradient, precise attenuation, inconclusive, reversal—covers the pooled within-GWA estimate cleanly. Its confidence interval excludes zero, so it is not inconclusive by the plan’s own definition (an interval containing both zero and the prior -2.10 estimate); but its coefficient of 0.99 narrowly misses the plan’s one-percentage-point materiality bar, so it does not qualify as a robust gradient either, and the interval’s lower end (-1.77) does not rule the bar out, so it does not qualify as a precise attenuation. This is a gap in the decision rule rather than a fifth outcome invented after the fact: the within-GWA association is negative, precisely estimated, about half the raw country-level magnitude, and just short of the threshold that was meant to separate a robust gradient from an attenuated one.

Older vintages retain the composition decomposition because no direct country–task–mode panel exists. Writing s_{ik} for country i ’s share of usage on category k , a_k for that category’s global automation share, and A_i for country i ’s published automation share, define

$$\hat{A}_i = \sum_{k \in K_i} \frac{s_{ik}}{c_i} a_k, \quad c_i = \sum_{k \in K_i} s_{ik}, \quad R_i = A_i - \hat{A}_i, \quad (60)$$

where K_i is the set of published categories carrying a global automation rate. $\hat{A}_i$ is the automation

share implied by country i 's task mix at global within-category modes, and R_i is the remainder. The full-release decompositions cover independent cross-sectional cohorts of 114–116 countries in August, November, and February—not a balanced panel, since the three snapshots share no common country set. Their predicted-composition gradients are -2.23 , -1.43 , and -1.40 , against observed gradients -4.16 , -0.74 , and -3.08 . The validation exercise computes the same decomposition where the direct GWA regression is available. In April and May, predicted-composition gradients are slightly positive ($+0.10$ and $+0.06$) while aggregate gradients are -2.03 and -2.24 ; the pooled direct within-GWA coefficient is only -0.99 , and the change estimates are positive and imprecise. Thus the decomposition correctly says that broad GWA mix does not generate the cross-sectional aggregate gradient, but it does *not* recover the direct conditional coefficient. That disagreement limits how strongly the older-vintage residuals can be read.

Table 2 reports both the connector-era country robustness battery and the release-data estimand ladder; Table 3 adds the same-cohort validation rows. The raw coursework gradient remains a composition fact rather than a depth measure: adding published category shares moves it from -5.17 (0.55) to $+0.15$ (0.62), while the direct work-use share carries no robust gradient. The new release panel changes the evidentiary status of Fact 3 in the opposite direction: it supplies a negative within-GWA cross-section, but not a precise within-country change result.

Two measurement issues cut across the whole battery, and robust standard errors address neither. Shares are published to two decimal places, cells that fail the aggregation and geographic sample floors are censored, and a missing cell means unpublished rather than zero, so effective precision differs enormously between high- and low-volume countries: the median published country accounts for 0.16 percent of observed world usage, 37 percent of them for less than 0.1 percent each. Hence the volume-weighted and high-volume specifications, which are not sensitivity checks but different estimands—the second is the population of countries measured precisely enough for the shares to mean much. And the sample is countries clearing publication thresholds, together 87.5 percent of observed world usage: selection is on the very intensity that is the regressor, which biases the sample toward the upper part of the distribution and is a reason to read the coefficients as within-cohort orderings rather than global gradients.

Frontier traffic against the ladder's new rungs. The task-level 1P API shares, mapped to O*NET with mode splits, are the raw material for a displacement-exposure index for digitally delivered service exports. The release files now provide the marginals directly. Record-weighted top-ten task concentration is 37.3 percent in August 2025, 41.6 in November, and 40.0 in February 2026; completion-token weighting gives 48.2, 42.5, and 38.6 percent, and cost weighting 51.9, 51.1, and 50.0 percent. The report's 28-to-33 comparison is therefore not reproduced by the available record-weighted releases: the August value is 37.3, and the nearest later released snapshot is 40.0; differences in sample window, task definition, or weighting are the likely reconciliation and prevent substituting one series for the other.¹ Among classified legacy records, the directive-plus-feedback

¹The June record-weighted top-ten shares are 23.9 and 24.1 percent in April and May, but the June classifier and product universe constitute a known break: the February API sample includes Claude Code and June excludes it. They are reported side by side, never as a common trend.

share is 86.2, 83.9, and 79.7 percent; the June automation bucket is 93.7 and 94.2 percent, again not comparable in level across the break. Those task families overlap with software and IT services, back-office and finance-and-accounting work, and contact-centre and business-process services in India and the Philippines. That remains broad task overlap, not exposure: the API feed is one vendor’s own traffic and measures what its customers ask the model to do, not what world demand delegates to the frontier.

The adopter side cannot close that gap either. Indian and Philippine chat profiles concentrate on the same rungs, but only barely relative to the world: the export-relevant triple of Computer and Mathematical, Office and Administrative Support, and Business and Financial Operations accounts for 39.1 and 38.9 percent of own usage against a 37.5 percent world baseline, and within it Computer and Mathematical runs at 1.10 and 1.12 times baseline, the software-development request topic at 16.2 and 16.6 percent against 11.5 worldwide, with the Philippine profile adding the creative-services signature of its outsourcing industry (arts, design and media at 1.40 times baseline; client copy-editing at 1.6 times). Conversations are matched to occupational tasks, not users to occupations, so these profiles could reflect students, domestic developers, hobbyists, or local firms as easily as workers in export industries. The defensible interim conclusion is therefore narrow: the public data show substantial overlap between the task families prominent in frontier API traffic and those associated with IT and business-service activity in two prominent developing-country exporters of digitally delivered services, which makes digitally delivered services the natural empirical setting for the model; they cannot attribute API adoption to competitor countries, link chat users to export-sector employment, or establish that the effect is quantitatively large. Whether a large a_P on these rungs destroys or expands employment remains the world-demand question ζ_P of prediction P4—and the size of a_P itself remains to be estimated rather than read off a usage share.

The index that would sharpen this is the exporter-level version of the aggregation (21) introduced with the microfoundation: for exporter i , $\text{Exposure}_i = \sum_k \omega_{ik} \pi_k^{\text{API}}$, with ω_{ik} the share of country i ’s digitally delivered service exports accounted for by activities mapping into task family k and π_k^{API} that family’s computed share of mapped API traffic, separately by collabouration mode and weighting unit. The O*NET-level API input is now computed; the binding missing component is the external EBOPS/BaTiS-to-task crosswalk. A complementary same-release comparison matches 8,241 June task-family cells across product surfaces: the API automation share exceeds chat/Cowork by 36.2 percentage points on average. This is a matched-*label* surface gap, not the same work under two treatments: users, applications, task complexity, and call structure differ, so it is suggestive evidence that deployment context moves measured collabouration mode and not an estimate of d_s .

What the public data cannot do—yet. Stage 0 verifies, file by file, that geography exists for chat products and is absent from the public 1P API aggregates. The buyer-side shift-share of Section 9.2 therefore cannot be run at country level: task composition is measurable, attribution to the buyer countries whose demand it serves—the attribution the foreign-competition component ν_s needs—is not. That is this paper’s sharpest data request: buyer-country–task–mode aggregates for API traffic, at whatever coarseness clears privacy review. The remaining structural inputs are

external to any usage index by construction— ψ_s from pre-AI variation, ζ_s and μ_s from trade data, χ_C from expenditure surveys, and sectoral weights from employment statistics. The exploratory GWA crosswalk cannot fill the last gap: depending on where ambiguous occupational families are placed, the global ratio of C -associated to P -associated observed use spans 0.02–1.22. It therefore neither measures a_C nor challenges or supports the maintained benchmark $a_C = 0$. The full releases make the AEI block reproducible and multi-vintage; they still cannot discipline the development multiplier.

Dependent variable and specification	Slope on ln AUI	Robust se	Clustered
<i>Automation-style share of conversations (pp)</i>			
(1) unweighted	−2.10	0.18	0
(2) + job-category composition	−1.93	0.26	0
(3) weighted by usage volume	−1.58	0.50	0
(4) high-volume half of sample	−2.14	0.29	0
(5) composition-adjusted and weighted	−2.68	0.31	0
(6) excluding imputed category shares	−2.00	0.19	0
(7) as (6), composition-adjusted	−1.74	0.27	0
leave-one-out range, (1) and (2)	[−2.15, −2.04] and [−2.05, −1.86]		
<i>Release-data within-GWA estimand ladder, Apr.–May 2026</i>			
equal-country, GWA + month FE, controls	−0.99	—	0
equal-cell, GWA + month FE, controls	−0.98	—	0
traffic-weighted, GWA + month FE, controls	−2.13	—	0
equal-country + country FE	+0.41	—	0
first difference	+0.71	—	0
country-aggregate change (full 114-country cohort)	−3.02	1.23	0
<i>Pre-specified selection diagnostics, same specification and controls</i>			
high-coverage countries (≥80% of observed GWA universe)	−0.34	—	0
common-GWA support (13 GWAs published by every balanced country-month)	−1.10	—	0
<i>Coursework share of conversations (pp)</i>			
(1) unweighted	−5.17	0.55	0
(2) + job-category composition	+0.15	0.62	0
(3) as (2), excluding education category	+0.16	0.62	0
(4) + January-start academic-year dummy	−4.95	0.54	0
(5) weighted by usage volume	−4.85	1.45	0
(6) high-volume half of sample	−4.67	0.96	0
leave-one-out range, (1)	[−5.31, −5.02]		
<i>Work-use share of conversations (pp)</i>			
(1) unweighted	+1.29	0.52	0
(2) + job-category composition	+0.32	0.49	0
(3) weighted by usage volume	−0.58	0.74	0

Table 2: Delegation gradients and their robustness. The first, coursework, and work-use blocks reproduce the archived May 2026 chat/Cowork cross-section: unweighted OLS unless stated, HC1 standard errors in column three and seven-region clustered errors in column four. The release-data block uses 76 countries with balanced GWA publication in April and May (median 20 of 35 GWAs published per country); controls are log 2024 GDP per capita and published work-use share, and column four reports two-way country/GWA clustered errors unless stated. The country-aggregate change row instead uses the full 114-country cohort publishing both months, with HC1 (column three); it is a country-level average, not a within-GWA result. Equal-country weights normalize the published GWA cells within each country to one; equal-cell and traffic weights target different populations. The country-FE row drops the time-invariant income control; the first-difference row contains one change per country–GWA. The selection-diagnostic rows are pre-specified in the analysis plan and use the same specification and controls as the primary row on the stated subsample. The primary −0.99 estimate has a 95 percent CI of [−1.77, −0.22] and Webb wild-GWA bootstrap $p = 0.017$; it narrowly misses the plan’s one-percentage-point materiality bar and its interval does not rule that bar out, a case the plan’s four-outcome decision rule does not cover (Section 9.3). Descriptive equilibrium-outcome relationships, with no causal content; missing cells are unpublished rather than zero.

Country	AUI	Observed A_i	Predicted $\hat{A}_i$	Residual R_i	Coverage (%)
Australia	6.40	45.50	47.59	-2.09	84.6
Singapore	5.81	46.30	48.23	-1.93	85.3
Switzerland	5.02	44.15	48.12	-3.97	85.6
New Zealand	4.84	46.33	47.82	-1.49	84.8
Canada	4.13	47.33	47.93	-0.60	85.1
Norway	4.04	41.26	47.15	-5.89	84.7
France	3.97	45.60	47.37	-1.77	85.4
Netherlands	3.90	45.27	48.74	-3.47	84.9
United States	3.87	49.59	48.04	+1.55	84.7
Belgium	3.80	42.70	47.19	-4.49	85.7
Korea, Rep.	3.78	43.76	48.20	-4.44	85.1
Denmark	3.57	42.27	47.87	-5.60	85.4
United Kingdom	3.35	44.81	46.98	-2.17	85.2
Portugal	3.32	47.49	49.02	-1.53	85.0
Germany	2.40	45.22	48.57	-3.35	85.5
Japan	1.91	47.45	49.12	-1.67	84.9
Brazil	0.96	51.18	48.82	+2.36	84.7
Bolivia	0.56	51.56	50.55	+1.01	71.5
Indonesia	0.46	47.51	47.39	+0.12	84.4
Philippines	0.44	49.17	48.47	+0.70	85.3
Kenya	0.34	54.61	48.31	+6.30	83.3
India	0.30	51.87	48.03	+3.84	86.0
Nigeria	0.24	51.70	45.38	+6.32	86.9
<i>Gradient on $\ln$ AUI (HC1 standard errors in parentheses, $N = 23$)</i>					
observed A_i	-2.47	(0.41), $R^2 = 0.60$; full 121-country sample	-2.10	(0.18)	
composition $\hat{A}_i$	+0.00	(0.27), $R^2 = 0.00$			
within-occupation R_i	-2.47	(0.41), $R^2 = 0.63$; leave-one-out	[-2.63, -2.23]		
<i>Same decomposition at job-category resolution, all 121 published countries</i>					
observed A_i	-2.10	(0.18), $R^2 = 0.54$			
composition $\hat{A}_i$	-0.05	(0.09), $R^2 = 0.00$			
within-category R_i	-2.05	(0.18), $R^2 = 0.54$; leave-one-out	[-2.10, -1.98]		
<i>Release-data validation at GWA resolution, balanced 76-country cohort</i>					
April: observed / composition / residual	-2.03	(0.27) / +0.10 (0.08) / -2.13	(0.25)		
May: observed / composition / residual	-2.24	(0.28) / +0.06 (0.08) / -2.30	(0.26)		
direct pooled within-GWA	-0.99	(0.40), 95% CI [-1.77, -0.22]; country FE	+0.41	(1.88)	

Table 3: Standardized decomposition of the delegation gradient and its release-data validation. $\hat{A}_i$ and R_i are defined in (60); since $A_i \equiv \hat{A}_i + R_i$, the three gradients sum exactly. The country rows and first block retain the archived connector-era occupation standardization for 23 selected countries; the next block repeats it at job-category resolution for all 121. The final block applies the same arithmetic at GWA resolution to the 76 countries balanced across April and May and compares it with the directly observed within-GWA regression. The decomposition attributes essentially none of the aggregate gradient to broad GWA mix, but the direct conditional coefficient is about half the residual gradient and the country-FE estimate reverses sign imprecisely. Agreement on the sign of composition does not license treating older-vintage residuals as direct within-task estimates. Author’s calculations; descriptive, with no causal content.

9.4 Predictions

With $\hat{\psi}_s$ estimated pre-AI and $\hat{\vartheta}_{is}$ from AI-era variation, the model delivers four falsifiable predictions.

P1 *Threshold interaction* (Propositions 4 and 5). *Fixed-wage* labour demand in tradable sector s turns positive above the capability threshold $\underline{H}_s$ of Proposition 4, so the interaction of sectoral AI exposure with absorptive capacity is positive above that threshold and negative below it. *Observed* sectoral employment is a different object. By (40) it rises only where $\vartheta_s > \mu\Lambda\Delta + \mu a_C$, after netting out the common general-equilibrium wage response; under the common- μ benchmark that displacement term is identical across tradable sectors within a country and period, so country–time fixed effects absorb it and the *relative* sectoral prediction is the sharp test, with the non-tradable service share absorbing the residual. First, with heterogeneous residual demand elasticities the observed response is $\partial_F \ln L_{ist} = \vartheta_{ist} - \mu_s \partial_F \ln w_{it}$ (Proposition 12), so the wage component varies across sectors through μ_s and a country–time effect no longer removes it; the relative test then requires either sector-specific $\hat{\mu}_s$ or an explicitly estimated wage term. Second, even under the common- μ benchmark the fixed effect identifies the relative sectoral response and not the absolute point at which $\vartheta_s = 0$: recovering the fixed-wage threshold requires the wage term itself, or the price, market-share, and demand structure of Section 9.2. Market shares behave differently again: by Proposition 2 the market-share response is weakly negative at every capability level and converges to zero only as $g(D) \rightarrow 1$. That is a limiting zero-loss condition—never attained at finite readiness under the maintained specification $g < 1$ —rather than a threshold at which a sign changes, so market shares should not exhibit the sign-switching interaction at all.

P2 *Covariance, not average* (Theorem 1). Conditional on average AI exposure, future capability growth loads on the *alignment* between sectoral exposure gains and ladder weights. The test is that the out-of-sample construction $\hat{\mathcal{M}}_{it}^c$ predicts capability growth conditional on—and better than—average exposure, average adoption, initial income, digital readiness, and initial sectoral composition, with split-sample or leave-one-country-out construction guarding against $\hat{\mathcal{M}}^c$ being fitted to the outcome it is meant to explain, and with the first-stage objects re-estimated inside each replication (Section 9.1). P2 is a test of the *composition* channel. It bears on the full development multiplier only under the maintained condition that the learning-content term of Theorem 1 is zero, separately controlled, or orthogonal to $\hat{\mathcal{M}}^c$ —a condition to be stated, not assumed silently, since that term can reverse the sign of the total effect.

P3 *Aggregate invisibility as a dynamic sequence* (Proposition 7). The prediction is an event-study *ordering*, not a cross-sectional correlation: on impact, wages, real consumption, and consumer-service employment rise in economies whose composition places them in the trap region; with a lag, capability-sensitive outcomes (export sophistication, upgrading, entry into complex activities) deteriorate. Because the post-2022 window is short relative to capability dynamics, the honest implementation combines high-frequency evidence on the

impact stage with historically estimated reallocation-to-capability relationships for the lagged stage, keeping directly observed and model-implied effects clearly separated. The classification is a *composition*-trap classification unless the learning-content term is proxied or sign-restricted, so “trapped” should be used in that qualified sense throughout. And because every first-stage input is estimated with error, the test should enter $\widehat{\mathcal{M}}^c$ continuously—interacting it with the AI shock—rather than splitting countries into trapped and non-trapped groups on noisy plug-in point estimates.

P4 *World demand elasticity* (Propositions 1 and 2). In sectors with price-inelastic world demand ($\zeta_s \leq 1$), frontier AI shocks reduce *fixed-wage* home labour demand at every level of absorptive capacity—no capability threshold exists—whereas threshold behavior appears only where $\zeta_s > 1$; market shares, by contrast, respond to absorption in both regimes. As in P1, the prediction is about ϑ_s rather than about raw employment: if $\bar{\vartheta} < 0$ the induced wage decline in (40) can leave *observed* employment rising in the least affected tradable sector even though every $\vartheta_s < 0$, so the test must be run on employment net of a country–time effect. In principle ζ_s is estimable from trade data without AI data, but the requirement is demanding: ζ_s (world demand for the sector) must be separated from μ_s (substitution across supplying countries), and from quality change, new products, and destination income growth. P4 is a structural demand-estimation exercise, not a free out-of-sample test.

9.5 Where to look first

The sector where the optimistic and pessimistic narratives collide is tradable services: business-process outsourcing, customer support, software and IT-enabled services, translation and content work, accounting and back-office services, online freelancing. These activities are simultaneously the candidate new rungs of the ladder (ψ_P large, on the optimistic reading), a natural candidate for substantial exposure to generative AI—Fact 2 of Section 2 identifies the relevant task families, while the magnitude of a_P remains to be estimated through the aggregation (21), since a_P is a structural parameter and not a usage share—and sold into world markets whose demand elasticity ζ_P is an open question. The disaggregation (1) matters most here: “tradable services” as listed above spans advanced software and professional work and routine transaction processing, and there is no reason to expect the two halves to share either ψ or ζ . A firm-, contract-, or worker-level study in one or several large exporters of such services—observing which tasks are automated or augmented, whether entry-level positions disappear, whether less experienced workers learn faster or stop learning (τ_P , η_P , and the mode share d_P in (25), which the learning response identifies only up to the frontier factor $\Psi'(F)$ common to all sectors, and which a firm-level design can separate precisely because it observes the mode of use and the learning outcome on the same tasks), and whether domestic value added rises while employment falls—identifies the model’s central objects more credibly than a global country-sector panel, and is the natural design for the companion paper. The remaining structural inputs follow the estimation strategies already cited: elasticities (μ_M, μ_P) separately (Appendix C shows what the common- μ baseline suppresses), the demand

block non-homothetically (the statistic χ_C of (32) is the exact object to estimate, from expenditure surveys in the manner of Fan et al. (2023), together with its relative-price component χ_C^q separately once $a_C \neq 0$, by Remark 12), and AI-shock magnitudes disciplined by task-level evidence in the manner of Acemoglu (2025).

10 Conclusion

The question of whether AI helps or hurts development is not a question about the size of the average productivity gain. In a minimal open-economy model of structural transformation with endogenous capability, the answer is an alignment condition with three parts: frontier AI raises a developing country’s capability growth when its labour-demand gains covary with sectoral learning weights, when the Baumol pull into local services—which operates wherever the frontier advance raises average tradable labour demand, and reverses where it does not—is weak, and when AI enriches rather than hollows out the learning content of the jobs that remain. The thresholds that govern the composition channel are emergent objects—they exist only where world demand is sufficiently elastic and readiness sufficiently high—and they are thresholds for fixed-wage labour demand rather than competitiveness: absorption always defends market share, yet it cannot defend a sector’s employment against inelastic world demand, because world expenditure on the sector falls whatever the home country does. In general equilibrium the observed employment response adds a common wage term (Proposition 5), so the threshold for observed employment sits at $\vartheta_s = \mu\Lambda\Delta + \mu a_C$, not at $\vartheta_s = 0$. First, readiness and productive capability are different stocks, and it is readiness that clears adoption thresholds while capability is what adoption is supposed to build. Second, how much AI is implemented and how it is implemented are different margins, and only the second determines whether the work that remains still teaches. Third, the Baumol penalty is driven by the AI shock to tradables *relative* to local services, so a maintained assumption that local services are unexposed is a choice about the sign of a first-order term rather than a simplification. The trap the alignment condition defines is invisible in aggregate indicators, welfare-relevant under a first-order and estimable condition, capable of generating discontinuous collapses and take-offs from continuous frontier progress, and priced, at the margin, by the multiplier itself. The framework does not assert that AI removes the development ladder. It provides the objects—ladder weights estimated from pre-AI variation, absorption separated from displacement, world demand elasticities, the exposure of local services, and the mode and learning content of AI-assisted work—for determining, country by country and sector by sector, whether AI raises domestic capability, commoditizes an existing rung, or redirects labour into low-learning services. Each of these statements is a restriction on data that existing sources can begin to test.

A Proofs

A.1 Proof of Proposition 10

Fix $c \in (0, c_0)$ where c_0 is small enough that $c_0 < r_{c_0}$ (possible since $r_\eta > 0$ for each η) and the tubes below are well defined.

Step 1 (tracking the upper branch). Let $\eta = c$ and consider the tube $\mathcal{T}_{\text{hi}} \equiv \{(H, F) : F \leq \hat{F} - c, |H - H_{\text{hi}}(F)| \leq r_c\}$, on which $\partial_H \mathcal{G} \leq -\kappa_c$ by Assumption 5(ii). Define $E_t = H_t - H_{\text{hi}}(F_t)$; while $(H_t, F_t) \in \mathcal{T}_{\text{hi}}$, the argument of Proposition 9 gives

$$|E_t| \leq e^{-\kappa_c t} |E_0| + \frac{\varepsilon B_c}{\kappa_c} (1 - e^{-\kappa_c t}).$$

With $E_0 = 0$, choosing $\bar{\varepsilon} \leq \kappa_c \min(c, r_c)/(2B_c)$ makes the right side at most $\min(c, r_c)/2$ for all t , so the trajectory can never reach the tube boundary before $F_t = \hat{F} - c$; hence claim (i).

Step 2 (transit). For $t \in [(\hat{F} - c - F_0)/\varepsilon, t_c]$, i.e., while $F_t \in [\hat{F} - c, \hat{F} + c]$, the state remains in the invariant set $\mathcal{K}$, so $H_{t_c} \in \mathcal{K}$. If $H_{t_c} \leq H_{\text{lo}}(F_{t_c}) + c$ skip to Step 4.

Step 3 (descent). For $t \geq t_c$ we have $F_t \geq \hat{F} + c$, so by Assumption 5(iii), as long as $H_t \geq H_{\text{lo}}(F_t) + c$,

$$\dot{H}_t = \mathcal{G}(H_t, F_t) \leq -\gamma_c.$$

Meanwhile $H_{\text{lo}}(F_t)$ moves at rate at most εB ; for $\bar{\varepsilon} \leq \gamma_c/(2B)$ the gap $H_t - H_{\text{lo}}(F_t)$ decreases at rate at least $\gamma_c/2$, so the trajectory enters the set $\{H \leq H_{\text{lo}}(F) + c\}$ within time $T_c \equiv 2\bar{\mathcal{K}}/\gamma_c$, where $\bar{\mathcal{K}}$ is the length of $\mathcal{K}$.

Step 4 (capture by the lower branch). Let $E'_t = H_t - H_{\text{lo}}(F_t)$ and let τ be the first time (after t_c) with $E'_\tau \leq c$. Note $E'_\tau \geq -\varepsilon B/\kappa_c - o(1) > -c$ for $\bar{\varepsilon}$ small, since the trajectory enters from above with $\dot{H} \leq 0$ bounded and the lower tube is stable; more simply, at τ we have $|E'_\tau| \leq c \leq r_c$, placing the state in the lower tube of Assumption 5(ii), where $\partial_H \mathcal{G} \leq -\kappa_c$ uniformly in F . The Gronwall bound then gives, for $t \geq \tau$,

$$|E'_t| \leq e^{-\kappa_c(t-\tau)} |E'_\tau| + \frac{\varepsilon B}{\kappa_c} \leq \max\left(c, \frac{\varepsilon B}{\kappa_c}\right) \leq c$$

for $\bar{\varepsilon} \leq \kappa_c c/B$, and the trajectory remains in the tube thereafter. Since $\tau \leq t_c + T_c$, claim (ii) follows with the stated T_c . ■

A.2 Proof of Proposition 11

Write the current-value Hamiltonian of (56) with $T(\{L_s\}; H, F) = \sum_s \Xi_s(A_s L_s)^{1-1/\mu}$ and $C = A_C L_C$:

$$\mathcal{H} = \ln u(T, C) + \lambda \left(\sum_j \psi_j(H) L_j - \delta H \right) + \varpi \left(1 - \sum_j L_j \right).$$

Since $\partial T/\partial L_s = (1 - 1/\mu)\Xi_s A_s^{1-1/\mu} L_s^{-1/\mu} = (1 - 1/\mu)p_s A_s$ with $p_s = \Xi_s Y_s^{-1/\mu}$, the first-order conditions $\partial \mathcal{H}/\partial L_s = 0$ and $\partial \mathcal{H}/\partial L_C = 0$ are (57). The co-state equation is

$$\dot{\lambda} = \rho\lambda - \partial_H \mathcal{H} = \left(\rho + \delta - \sum_j \psi'_j(H) L_j\right)\lambda - \frac{u_T}{u} \left(1 - \frac{1}{\mu}\right) \sum_s p_s Y_s \partial_H \ln A_s,$$

which, under the standard transversality condition $\lim_{t \rightarrow \infty} e^{-\rho t} \lambda_t H_t = 0$ and provided the integral below converges—both maintained here rather than derived—integrates to

$$\lambda_t = \int_t^\infty \exp\left(-\int_t^\tau (\rho + \delta - \sum_j \psi'_j L_j) ds\right) \frac{u_T}{u} \left(1 - \frac{1}{\mu}\right) \sum_s p_s Y_s \partial_H \ln A_s d\tau \geq 0,$$

where the sign follows because every factor of the integrand is nonnegative, and is strict whenever the integrand is positive on a set of positive measure.

For decentralization, consider a competitive equilibrium in which producers in tradable sector s receive $(1 - 1/\mu)p_s$ per unit (an ad valorem export tax at rate $1/\mu$) plus an employment subsidy $\tau_s = \tilde{\lambda}\psi_s(H)$ per worker, and producers in C receive qA_C plus $\tau_C = \tilde{\lambda}\psi_C(H)$, with revenues rebated lump-sum. Zero profits and free mobility equalize the full return to labour:

$$w = \left(1 - \frac{1}{\mu}\right)p_s A_s + \tilde{\lambda}\psi_s(H) = qA_C + \tilde{\lambda}\psi_C(H).$$

Household optimality gives $q = u_C/u_T$. Multiplying the planner conditions (57) through by u/u_T and setting $\tilde{\lambda} = \lambda u/u_T$, $w = \varpi u/u_T$ shows the two systems coincide; goods markets clear by construction of the budget constraint. Since a common level shift in $(\tau_j)_j$ is absorbed by w , only the differentials matter. ■

B General demand systems

The main text states Theorem 1 in terms of the statistic χ_C ; this appendix supplies the underlying demand systems. What matters for the Baumol term is not CES or homotheticity per se; it is that a higher wage—through relative prices, income, or both—expands the local-service employment share. Suppose local-service expenditure is described by a differentiable Marshallian share function

$$L_C = s_C(q, w), \quad q = \frac{w}{A_C}, \quad (61)$$

so that the statistic defined in (32) is

$$\chi_C = \chi_C^q + \chi_C^w, \quad \chi_C^q \equiv \partial_{\ln q} s_C(q, w), \quad \chi_C^w \equiv \partial_{\ln w} s_C(q, w), \quad (62)$$

the sum of a relative-price response and an income response.

Remark 12 (One statistic, or two). *Under the unexposed-services benchmark $a_C = 0$ the wage and the relative price move one for one, and only the sum χ_C is ever needed—which is why the main*

text states its results in terms of a single demand statistic. Once $a_C > 0$ that is no longer true: an A_C shock moves the relative price directly without moving household income directly, so the two components enter separately. Repeating the derivation of Proposition 5 with $\partial_F \ln q = \partial_F \ln w - a_C$ and $\partial_F L_C = \chi_C \partial_F \ln w - \chi_C^q a_C$ gives

$$\partial_F \ln w = \frac{L_T \bar{\vartheta} - \chi_C^q a_C}{\mu L_T - \chi_C}, \quad \partial_F \dot{H} = L_T \left[\text{Cov}_\xi(\psi_s, \vartheta_s) - \frac{(\bar{\psi} - \psi_C)(\chi_C \bar{\vartheta} - \mu \chi_C^q a_C)}{\mu L_T - \chi_C} \right] + \sum_j L_j \partial_F \psi_j. \quad (63)$$

Homotheticity sets $\chi_C^w = 0$, so $\chi_C^q = \chi_C$ and (63) collapses to (42). In the non-homothetic case relevant to developing economies, where income effects are the larger part of χ_C (Fan et al., 2023), $\chi_C^q < \chi_C$, so local-service exposure offsets less of the Baumol penalty than the homothetic formula implies: the two statistics must be estimated separately, and an expenditure-survey design that recovers only their sum is not sufficient once $a_C \neq 0$.

Static equilibrium requires a wage solving $\Omega w^{-\mu} = 1 - s_C(w/A_C, w)$; a sufficient condition for existence and uniqueness generalizing Proposition 3 is that $R(w) \equiv w^\mu [1 - s_C(w/A_C, w)]$ be continuous and strictly increasing from 0 to ∞ . Strict monotonicity holds whenever $\chi_C(w) < \mu L_T(w)$ along the equilibrium relations—the second part of Assumption 3 imposed globally—because $d \ln R / d \ln w = \mu - \chi_C / L_T > 0$ there. Monotonicity does not by itself deliver the endpoints, which must be imposed separately:

$$\lim_{w \downarrow 0} w^\mu [1 - s_C(w/A_C, w)] = 0, \quad \lim_{w \rightarrow \infty} w^\mu [1 - s_C(w/A_C, w)] = \infty. \quad (64)$$

The first holds whenever s_C is bounded. The second is the substantive one: it rules out demand systems in which the tradable expenditure share vanishes fast enough to offset w^μ , and it holds under CES because $1 - s_C \sim [\omega / (1 - \omega)]^\sigma q^{-(1-\sigma)}$ with $\mu > 1 - \sigma$, as in the proof of Proposition 3. Given (61), the derivations of Proposition 5 and Theorem 1 in the main text apply verbatim under the benchmark $a_C = 0$: differentiating (61) at fixed states, with $\partial_F \ln q = \partial_F \ln w$, gives $\partial_F L_C = \chi_C \partial_F \ln w$, and labour-market clearing $0 = L_T(\bar{\vartheta} - \mu \partial_F \ln w) + \chi_C \partial_F \ln w$ delivers the pass-through $\Lambda = L_T / (\mu L_T - \chi_C)$ of (34). Remark 12 gives the general case.

Remark 13 (CES verification). Under (2), s_C depends only on q and $\partial_{\ln q} s_C = (1 - \sigma)s_C(1 - s_C)$ by (5), with no income term by homotheticity; using $s_C = L_C$ and $1 - L_C = L_T$ gives $\chi_C = (1 - \sigma)L_C L_T$, which is (33), and $\chi_C < \mu L_T$ holds because $\sigma \in (0, 1)$ and $\mu > 1$.

Remark 14 (Stone–Geary or PIGL demand). Under non-homothetic demand, χ_C combines a substitution and an income effect. In developing economies, where income effects dominate the expansion of consumer services (Fan et al., 2023), plausible calibrations make χ_C larger than its CES counterpart, magnifying the Baumol term—a larger pull into local services where $\bar{\vartheta} > 0$, and a larger push out of them where $\bar{\vartheta} < 0$. The theorem shows exactly what the trap logic requires: a wage-elastic local-service employment share in the relevant region. Empirically, χ_C is the object to

estimate—from household expenditure data, Engel curves, and relative prices—rather than the CES parameter σ .

C Heterogeneous residual demand elasticities

If tradable sectors face different residual demand elasticities,

$$p_s = \Xi_s(F)Y_s^{-1/\mu_s}, \quad \mu_s > 1, \quad \vartheta_s \equiv (\mu_s - 1)\alpha_s(H, D) - \mu_s\nu_s, \quad (65)$$

define $\bar{\mu} \equiv \sum_{s \in S} \xi_s \mu_s$. This appendix is stated under the unexposed-services benchmark $a_C = 0$; carrying a_C as well replaces $\bar{\vartheta}$ by $\bar{\vartheta} - \bar{\mu}a_C$ in (66)–(67) and adds a term in $\text{Cov}_\xi(\psi_s, \mu_s)a_C$, without changing the structure of either.

Proposition 12 (Heterogeneous elasticities). *Under (65) and the generic demand system (61), and shutting down the learning-content channel ($\partial_F \psi_j \equiv 0$, which otherwise adds the same $\sum_j L_j \partial_F \psi_j$ term as in Theorem 1),*

$$\partial_F \ln w = \frac{L_T \bar{\vartheta}}{L_T \bar{\mu} - \chi_C}, \quad (66)$$

$$\partial_F \dot{H} = L_T \left[\text{Cov}_\xi(\psi_s, \vartheta_s) - \frac{\chi_C(\bar{\psi} - \psi_C)\bar{\vartheta} + L_T \bar{\vartheta} \text{Cov}_\xi(\psi_s(H), \mu_s)}{L_T \bar{\mu} - \chi_C} \right]. \quad (67)$$

Proof. labour demand is $L_s = \Xi_s^{\mu_s} A_s^{\mu_s-1} w^{-\mu_s}$, so $\partial_F \ln L_s = \vartheta_s - \mu_s \partial_F \ln w$ and $\partial_F L_T = L_T \bar{\vartheta} - L_T \bar{\mu} \partial_F \ln w$. Clearing against $\partial_F L_C = \chi_C \partial_F \ln w$ gives (66). Then

$$\begin{aligned} \partial_F \dot{H} &= \sum_{s \in S} (\psi_s - \psi_C) L_s (\vartheta_s - \mu_s \partial_F \ln w) \\ &= L_T [\text{Cov}_\xi(\psi_s, \vartheta_s) + (\bar{\psi} - \psi_C)\bar{\vartheta}] - L_T \partial_F \ln w [\text{Cov}_\xi(\psi_s, \mu_s) + (\bar{\psi} - \psi_C)\bar{\mu}], \end{aligned}$$

and substituting (66) and simplifying yields (67). $\square$

The common- μ baseline does not create the trap mechanism; it removes the elasticity-alignment term $\text{Cov}_\xi(\psi_s, \mu_s)$ —whether capability-building sectors also happen to face more elastic world demand—and thereby isolates the pure covariance between ladder weights and AI-induced labour-demand effects. A quantitative implementation should allow $\mu_M \neq \mu_P$ and estimate them separately.

D Alternative learning technologies

The baseline law (26) makes the worker-year the unit of learning: a sector contributes capability in proportion to its employment. AI is precisely a technology that lets fewer workers produce more output, serve more demanding clients, and perform more complex tasks, so the choice of learning basis is not innocuous—it can determine the sign of the development effect. This appendix derives

the multiplier under the polar alternative, output-based learning, and states the empirical moment that discriminates between the bases.

Suppose capability accumulates with sectoral production:

$$\dot{H} = \sum_{s \in S} \tilde{\psi}_s Y_s + \tilde{\psi}_C Y_C - \delta H, \quad (68)$$

with constant output-learning weights $\tilde{\psi}_j \geq 0$ (state-dependence adds nothing to the point).

Proposition 13 (Output-based multiplier). *Under (68) and Assumption 3,*

$$\partial_F \dot{H} = \sum_{s \in S} \tilde{\psi}_s Y_s (\alpha_s + \vartheta_s - \mu \Lambda \Delta - \mu a_C) + \tilde{\psi}_C A_C (a_C L_C + \chi_C \Lambda \Delta). \quad (69)$$

Proof. From $Y_s = A_s L_s$, $\partial_F \ln Y_s = \alpha_s + \partial_F \ln L_s = \alpha_s + \vartheta_s - \mu \Lambda \Delta - \mu a_C$ by (40); from $Y_C = A_C(F) L_C$, $\partial_F Y_C = A_C(a_C L_C + \partial_F L_C) = A_C(a_C L_C + \chi_C \Lambda \Delta)$ by (39). Differentiate (68). $\square$

Three differences from Theorem 1 follow. First, the direct productivity term α_s enters: a frontier advance can raise a sector's learning contribution while its employment falls, because output rises. At a fixed wage the sector's output effect is $\alpha_s + \vartheta_s = \mu(\alpha_s - \nu_s)$, which under the microfoundation equals $a_s[\zeta_s - \mu(1 - g(D))]$: the output threshold solves $g(\underline{D}_s^Y) = 1 - \zeta_s/\mu$, strictly below the employment threshold $(\mu - \zeta_s)/(\mu - 1)$ of (24), and—crucially—it can be interior when $\zeta_s \leq 1$, which the employment threshold never is. Precisely, it is interior when $0 < \zeta_s < \mu$ and $g(0) < 1 - \zeta_s/\mu$; if $g(0) \geq 1 - \zeta_s/\mu$ the output response is already positive at zero readiness. The boundary case is instructive: at $\zeta_s = 0$ the condition reads $g = 1$, which no finite readiness level attains, so fixed-wage output falls at every finite D and merely approaches a zero response under full absorption. Subject to that qualification, the “pure displacement” regime of Proposition 1, in which no absorption level helps, *does not exist* under output-based learning: for any $\zeta_s > 0$ there is a finite readiness level above which a frontier advance raises the sector's fixed-wage output. Second, the Baumol term reverses sign: reallocation into C raises C output, so if local services carry any output-learning weight $\tilde{\psi}_C > 0$, the term $\tilde{\psi}_C A_C \chi_C \Lambda \Delta$ is a Baumol *bonus*, not a drag (and AI inside local services adds a further bonus $\tilde{\psi}_C A_C a_C L_C$); the drag survives only through the $-\mu \Lambda \bar{\vartheta}$ compression of tradable output, and reallocation is no longer zero-sum. Third, the trap region shrinks accordingly: economies that are trapped under labour-based learning can be escalating under output-based learning at the same parameters.

Intermediate bases interpolate. If $\dot{H} = \sum_j \psi_j(H, F) \Upsilon_j(L_j, Y_j) - \delta H$ with Υ_j homogeneous of degree one and labour elasticity $\epsilon_j \equiv \partial \ln \Upsilon_j / \partial \ln L_j \in [0, 1]$, then using $\partial_F \ln Y_j = \partial_F \ln L_j + \alpha_j$,

$$\partial_F \dot{H} = \sum_j \psi_j \Upsilon_j [\partial_F \ln L_j + (1 - \epsilon_j) \alpha_j] + \sum_j L_j \partial_F \psi_j \cdot \frac{\Upsilon_j}{L_j},$$

so the entire difference between the bases is the weight $(1 - \epsilon_j)$ placed on the direct productivity gain α_j ; learning attached to exports or buyer relationships (Atkin et al., 2017) behaves like the

output pole insofar as export volumes track output.

The bases are empirically distinguishable, and Section 9.1 builds the test in: under labour-based learning, employment composition predicts future capability *conditional on* output, productivity, and export scale; under output-based learning the conditional effect vanishes and scale variables carry the load. Which pole the data favor determines whether the Baumol reallocation at the heart of the trap is a real development cost or an accounting artifact of the wrong learning basis—which is why the estimation of ψ must precede, and not be fitted jointly with, the AI-era analysis.

E Readiness dynamics: the planar system

The main text carries productive capability H and AI readiness D as two predetermined stocks and collapses them to one only where the transition analysis needs a one-dimensional state, through the diagonal convention (19). This appendix supplies what that convention stands in for: the accumulation law for readiness, the planar dynamics, and the conditions under which the one-dimensional analysis of Section 7 is a legitimate reduction of them.

Let H retain the learning-by-doing law (26), and let readiness evolve as

$$\dot{D} = \iota - \delta_D D, \quad \iota \geq 0, \quad \delta_D > 0, \quad (70)$$

where ι is investment in digital infrastructure, compute access, licensing, and complementary technical skills—a policy choice, or the solution of a separate accumulation problem whose determination is not needed here. The two laws differ in exactly the way the motivating evidence suggests they should: H moves only through the allocation of labour across sectors and therefore at the pace of learning by doing, while D responds directly to investment and can move quickly, including from outside the country when a product is launched, localized, repriced, or licensed into a new market.

Proposition 14 (The sufficient statistic is a statement about the pair). *For every (H, D, F) the static equilibrium exists and is unique, and $\partial_F \dot{H}$ is given by (42) evaluated at that state. No result in Sections 4–8 uses (19) or any other relation between H and D .*

Proof. No step in Proposition 3, Proposition 5, or Theorem 1 used the functional dependence of α_s on the states; the static block requires only the values α_s and hence Ω_s , with H and D entering as parameters of the period economy. $\square$

Alignment is genuinely two-dimensional. The covariance in (42) pairs a capability object, $\psi_s(H, F)$, with a readiness object, $\vartheta_s(H, D)$ —which under the microfoundation (23) depends on D alone. A country can therefore be AI-ready and capability-poor: it clears every threshold of Proposition 1, defends its market shares, and still accumulates little, because the labour-demand gains land in sectors with low ψ_s . It can equally be capability-rich and unready, so that the alignment question does not arise because nothing is implemented. In the one-stock model these are the same point on a line.

The dynamics are planar, and the baseline is a section of them. The system is $(\dot{H}, \dot{D}) = (\mathcal{G}(H, D, F), \iota - \delta_D D)$ with $\dot{F} = \varepsilon$, and the trap region of Definition 1 is a region of the (H, D) plane rather than an interval. Two reductions recover Section 7. If ι is set to hold D at $d(H)$ —readiness policy indexed to the level of development, which is roughly what a constant investment rate delivers—the system reduces exactly to the diagonal convention (19). If instead δ_D and ι are small relative to the learning dynamics, D is quasi-fixed over the horizon of a capability transition and the analysis of Section 7 applies on each D -slice, with the fold structure of Section 7.2 holding along any path on which D moves slowly. What the reductions rule out is the case the readiness evidence makes most salient: fast movements in D , which shift the entire drift field on a timescale short relative to capability accumulation and which can carry an economy across a fold without any change in its capability at all.

Two stocks, two instruments. For policy the split is operational rather than cosmetic. Investment ι builds the stock that clears the adoption thresholds of Proposition 1; the ladder subsidies of Section 8 steer labour toward the sectors that convert adoption into capability. Neither is sufficient alone, and they are not substitutes: readiness policy moves the economy up g , raising α_s and hence ϑ_s in every exposed sector at once, which is precisely the shock that Corollary 1 shows generates pure Baumol reallocation when the tradable sectors are dynamically alike. A readiness push evaluated on its own terms—adoption achieved, market shares defended—can therefore be a development loss, and the multiplier is the statistic that says which.

F Two labour types: a skill barrier to progressive services

The baseline has homogeneous labour, a common wage, and frictionless mobility—restrictive assumptions for the paper’s central reallocation claim, because the workers AI displaces from manufacturing are rarely the workers who could staff progressive tradable services, and the workers who can are a scarce minority. This appendix introduces the minimal heterogeneity: a fraction $\theta \in (0, 1)$ of the workforce is *skilled* and can work in any sector, while the remaining $1 - \theta$ are *unskilled* and can work only in manufacturing and local services. Within any sector the two types are perfect substitutes, production and demand are as in the baseline (CES, for concreteness; the general- χ_C treatment is analogous), and P employs skilled labour only. Let w_k and w denote the skilled and unskilled wages.

There are two regimes, and the baseline is one of them. If skilled labour is abundant enough that some skilled workers are employed outside P , within-sector substitutability forces $w_k = w$ and the homogeneous-labour model of the main text applies verbatim: the baseline is the *slack* regime, not a different model. The economically new case is the *binding* regime, $w_k > w$, in which all skilled workers are in progressive services:

$$L_P = \theta, \quad w_k = \left(\frac{\Omega_P}{\theta} \right)^{1/\mu}, \quad L_M = \Omega_M w^{-\mu}, \quad L_M + L_C = 1 - \theta,$$

with $q = w/A_C$, total income $I = \theta w_k + (1 - \theta)w$, and C -market clearing $wL_C = s_C(q) I$. Define

the skilled income share and the (I -based) local-service expenditure share

$$e \equiv \frac{\theta w_k}{I} \in (0, 1), \quad s_C = \frac{w L_C}{I},$$

and assume the binding-regime analogue of Assumption 3,

$$\Delta_T \equiv \mu L_M + e L_C - (1 - \sigma)(1 - s_C) L_C > 0, \quad (71)$$

which holds in particular whenever $\mu L_M \geq (1 - \sigma) L_C$.

Proposition 15 (Two-type multiplier in the binding regime). *In the binding regime, a marginal frontier advance moves equilibrium objects according to*

$$\partial_F \ln w_k = \frac{\vartheta_P(H)}{\mu}, \quad \partial_F L_P = 0, \quad (72)$$

$$\partial_F \ln w = \frac{L_M \vartheta_M(H) + e L_C \vartheta_P(H)/\mu}{\Delta_T}, \quad (73)$$

$$\partial_F \dot{H} = (\psi_M - \psi_C) L_M (\vartheta_M(H) - \mu \partial_F \ln w) + \sum_{j \in S \cup \{C\}} L_j \partial_F \psi_j(H, F), \quad (74)$$

with ψ_j evaluated at (H, F) .

Proof. Inverting $\theta = \Omega_P w_k^{-\mu}$ gives $w_k = (\Omega_P/\theta)^{1/\mu}$ and hence $\partial_F \ln w_k = (1/\mu) \partial_F \ln \Omega_P = \vartheta_P/\mu$ by (18), with $L_P = \theta$ fixed; this is (72). For the unskilled block, write $\hat{w} \equiv \partial_F \ln w$. From $L_M = \Omega_M w^{-\mu}$, $\partial_F \ln L_M = \vartheta_M - \mu \hat{w}$. From $\ln L_C = \ln s_C(q) + \ln I - \ln w$ with $\partial_F \ln q = \hat{w}$, $d \ln s_C / d \ln q = (1 - \sigma)(1 - s_C)$ by (5), and $\partial_F \ln I = e \partial_F \ln w_k + (1 - e) \hat{w}$,

$$\partial_F \ln L_C = (1 - \sigma)(1 - s_C) \hat{w} + e \left(\frac{\vartheta_P}{\mu} - \hat{w} \right).$$

Unskilled market clearing, $\partial_F L_M + \partial_F L_C = 0$, then gives

$$L_M (\vartheta_M - \mu \hat{w}) + L_C \left[(1 - \sigma)(1 - s_C) \hat{w} + e \left(\frac{\vartheta_P}{\mu} - \hat{w} \right) \right] = 0,$$

and solving for $\hat{w}$ yields (73) with Δ_T as in (71). Differentiating (26) with $\partial_F L_P = 0$ and $\partial_F L_C = -\partial_F L_M$ gives (74). As $\theta \rightarrow 0$ (so $e \rightarrow 0$, $L_C \rightarrow s_C$), (73) reduces to the baseline pass-through $\Lambda \bar{\vartheta}$ of (38) for an economy whose only tradable is M . $\square$

The alignment margin is severed by the skill constraint. In Theorem 1, a labour-demand gain concentrated in the high- ψ sector was the escalator scenario. Here $\partial_F L_P = 0$: AI shocks to progressive services are absorbed entirely by the skilled wage, and the composition channel operates only through sectors open to unskilled labour. The covariance term of the baseline degenerates to the M -versus- C margin in (74).

Corollary 3 (Leapfrog without a pipeline). *In the binding regime, consider a pure progressive-services shock: $\vartheta_P > 0$, $\vartheta_M = 0$, $\partial_F \psi_j \equiv 0$. Then $\partial_F \ln w_k > 0$, $\partial_F \ln w > 0$, $\partial_F L_C > 0$, and*

$$\partial_F \dot{H} = -(\psi_M - \psi_C) L_M \mu \frac{e L_C \vartheta_P / \mu}{\Delta_T} < 0 \quad \text{whenever } \psi_M > \psi_C.$$

The shock that the baseline model could score as the escalator—AI raising labour demand exactly in the high-learning tradable-services sector—strictly *reduces* capability growth when the skill pipeline binds: the gains arrive as rents to inframarginal skilled workers, whose spending pulls unskilled labour out of manufacturing into local services. Aggregate income, both wages, and consumer-service employment all rise, so the aggregate invisibility of Proposition 7 survives labour heterogeneity in this configuration; what changes is the verdict, since the same observables now accompany an unambiguous capability loss. Whether the pipeline binds is thus itself a first-order empirical object, and it maps to the readiness stock of Appendix E: policies that expand θ (skills, training) reopen the alignment margin that the constraint closes.

The trap can become distributionally visible. From (72)–(73), the skill premium responds as $\partial_F \ln(w_k/w) = \vartheta_P/\mu - \partial_F \ln w$, which takes either sign—AI can widen or compress the skill gap depending on where its labour-demand effects land. More sharply, if AI simultaneously erodes manufacturing and boosts progressive services ($\vartheta_M < 0 < \vartheta_P$), the unskilled wage *falls* whenever $L_M |\vartheta_M| > e L_C \vartheta_P / \mu$, even while the skilled wage, aggregate income, and consumer-service employment rise. In that configuration the trap is no longer invisible: it announces itself contemporaneously in falling unskilled earnings and a widening premium—exactly the distributional signature that the representative-agent welfare statements of Section 6 average away, and a warning that a rising mean wage does not establish that the workers leaving tradables gain.

Informality fits inside the same structure. Reading unskilled C -work as the informal sector, the binding regime delivers the familiar pattern—displaced manufacturing workers absorbed into low-earning local services while the modern service sector booms with a scarce skilled workforce—without any mobility friction beyond the skill barrier itself. A richer treatment would add underemployment or an earnings penalty within C ; the two-type structure is the minimal version in which “who can climb” and “who gets displaced” are different people, which is the margin the baseline suppresses.

G Additional boundaries

Remark 15 (Capital embodiment). *If AI adoption is embodied in capital, replace $\alpha_s(H, D)$ by $\alpha_s(H, D, K)$ with $\partial_K \alpha_s > 0$, where K is domestic AI-capital availability. The H – K complementarity typically sharpens thresholds—low-capability countries absorb little both because learning capacity and investment capacity are weak—while cheap imported AI capital relaxes them. The net effect is ambiguous ex ante; the labour-only model is a clean lower-dimensional core.*

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